Perceived Income Risk Across the Income Distribution*

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Abstract

In incomplete-market models, the risk an agent perceives is the risk an econometrician estimates from realised earnings. Direct data on perceptions violate this assumption, and do so systematically across the income distribution. Conditional on staying in the same job, workers at the bottom perceive about twice the wage risk of those at the top, hold that belief more persistently, and revise it more strongly after a shock. This is a gradient with no counterpart in the realised process I estimate from matched earnings spells. I fit a learning rule that reproduces its level, its persistence, and its response to realised shocks, and find perceived risk departing from the truth in two directions at once: the bottom over-perceives its transitory risk while the middle and the top under-perceive their permanent risk. I trace the gradient to how legibly pay separates into a permanent and a transitory part, a primitive I measure from the itemised pay components of the same records, rising from about an eighth of transitory pay variance at the bottom to over a third at the top. Embedded in a life-cycle buffer-stock economy, the measured beliefs open a precautionary-saving and marginal-propensity-to-consume wedge that a full-information calibration cannot generate, one that rises with income and concentrates at the top, where perceived permanent risk falls furthest below the truth.

1 Introduction

In an incomplete-market economy, how much risk a household believes it faces is a primitive of its consumption and saving behaviour. Holding expected income and preferences fixed, an agent who perceives more risk to her future earnings accumulates a larger precautionary buffer and spends less out of a windfall, whether because prudence in the utility function or an occasionally binding borrowing constraint induces precautionary saving. The size and the heterogeneity of idiosyncratic income risk are accordingly among the central inputs of the heterogeneous-agent models now standard in macroeconomics, where they shape the distribution of wealth, the distribution of marginal propensities to consume, and, through both, the transmission of aggregate shocks and of policy. But is the risk these models assign their agents the same as agents actually perceive, and if not, does the gap between the two fall evenly across households or pull them apart?

The common practice in this literature is to calibrate income risk indirectly. The econometrician estimates the cross-sectional dispersion of realised earnings in panel data, decomposes it into components of differing persistence (Meghir and Pistaferri, 2004), and hands the resulting vari-

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ances to the agent as the risk she perceives. Perceived risk is set equal to realised risk: the agent is assumed to know exactly the variances the econometrician recovers, and to know nothing the econometrician cannot. The assumption is convenient but strong, and direct data on perceptions now allow it to be tested.

Measured directly, perceived risk violates this assumption systematically across the income distribution. In the New York Fed's Survey of Consumer Expectations (SCE), which elicits from each respondent a density forecast of her own wage growth over the next twelve months, workers in households earning under \$50k report wage uncertainty about 1.8 times as large as those in households above \$100k, conditional on staying in the same job. They also hold their beliefs more persistently within person, and revise them more strongly after a realised wage shock. The realised wage process shows none of the three. Matched same-job spells in the Survey of Income and Program Participation (SIPP) carry no comparable gradient, and if anything realised risk rises mildly with income. The gradient is one of beliefs, and equating perceived with realised risk fails on all three margins, significantly on the level and the persistence and, in point estimates, on the response to shocks.

This paper documents these patterns, fits a learning rule that reproduces them, and traces them to how pay is delivered across the income distribution, in three steps. The first is descriptive. After Section 2 sets out the measurement, Sections 3 and 4 establish the belief gradient and its dynamics against a realised process that shows no counterpart, and so fix what the rule must reproduce.

The second step fits a belief-updating rule. The agent cannot separate her permanent and transitory shocks; each period she sees one squared pay change and must learn the variance of her permanent earning power from it. This is a signal-extraction problem in the tradition of Pischke (1995), set on the second moment instead of the level. In Section 5 I model it as an exponentially weighted average of past squared surprises, the steady-state Kalman filter for an unknown variance that follows a random walk. The rule has three interpretable parameters: how fast beliefs update, what share of each squared surprise the agent reads as news about permanent earning power, and a prior on transitory risk. Section 6 takes it to the data. Estimated within person, the bottom updates most slowly, reads about three times as much of each squared surprise as permanent as the top does, roughly twice the decline full information would imply, and holds a prior almost four times its realised transitory risk, while the middle and the top sit close to one another and near their realised processes. The persistence and prior gradients are statistically significant in the baseline; the attribution gradient is monotone but not.

The third step, in Section 7, asks why the deviations take the form they do, and here the paper departs from the standard model. The fitted rule reproduces the gradient without explaining it, since its three parameters are estimated freely group by group. What the rule leaves unexplained runs in two directions. Measured against the risk each group actually faces, the bottom over-perceives its transitory risk, while the middle and the top under-perceive their permanent risk. No one-sided friction can produce a cross-section that sits on both sides of the truth, so I look for two forces of opposite sign, and locate both in how pay is delivered.

The mechanism focuses on how legibly a worker's pay separates into a permanent and a transitory part. I measure this legibility directly, and it rises with income. At the top, transitory pay arrives labelled and against a single salary, so the worker nets it out before she updates and her perceived risk falls. At the bottom it arrives fragmented, as variable shifts, cash tips, and irregular supplements layered on uncertain hours, so she cannot separate the transitory part, and unresolved reading noise inflates her perceived risk instead. The SIPP itemises the components of a paycheck, and on the same stayer panel that supplies the realised process, the labelled share of transitory pay rises from about an eighth at the bottom to over a third at the top. This one feature of delivery sets the persistence of beliefs, their response to news, and the under-perception of permanent risk higher up. Where pay is legible the worker separates the transitory part cleanly and faces a sharper signal about the permanent part, which she then under-reads; the gradient in beliefs is a gradient in delivery.

Before running any simulation, I show that what the estimated mechanism implies for behaviour follows analytically. In Section 8 I show that the gap between an agent's perceived second moments and the true ones is a perceived-risk wedge. Because it is an error in a variance, it reaches saving through prudence, where a level error would act through the intertemporal-substitution motive. It bears on the perceived risk to permanent earning power, and hardly at all on the transitory prior, which a buffer absorbs within the period. In Sections 9 and 10 I embed the measured beliefs in a life-cycle buffer-stock economy in which workers drift across income groups as their permanent earning power accumulates, and in which a single discount factor, common across agents, is calibrated to average liquid wealth, so that the cross-section of wealth and marginal propensities is an outcome of the risk process. Relative to an otherwise identical full-information calibration, the beliefs open a precautionary-saving and marginal-propensity-to-consume (MPC) wedge that full information cannot itself generate. The saving wedge rises with income, small at the bottom and several times larger at the top under the measured shares, larger still at the upper edge of the measurement band and where the estimated belief responses are taken at face value. The behavioural wedge in the MPC concentrates at the top, whose permanent belief is discounted most, and liquidity reinforces this incidence without creating it. In one-asset general equilibrium the belief economy supplies modestly less capital at a slightly higher interest rate, so the wedge shows in the cross-section more than in the aggregates. The bottom holds the most distorted transitory prior of any group, yet its permanent belief is close to the truth, so its behavioural wedge is small.

The mechanism of Section 7 is measured at its primitive and scored against belief moments it does not fit, and its strongest test, independent variation in how pay is delivered, is out of sample and remains to be run. The model of Sections 9–11 is solved in partial and one-asset general equilibrium and traced through a recession and a risk shock in Section 11.

Related literature

The paper closest to this one is Wang (2023), who first used the SCE's density forecasts to show that perceived wage risk lies well below the risk conventionally calibrated from panel data, even within demographic groups, and that feeding this lower, more heterogeneous risk into an incomplete-market model raises wealth inequality and the hand-to-mouth share. I share his measurement strategy, the SCE density forecast for perceived risk set against the SIPP same-job stayer panel for the realised process, and I find his sign for the middle and the top, which at the twelve-month horizon perceive less risk than the full-information benchmark implies, a gap Section 4 reads as an upper bound. I depart from him in four respects. First, I organise the deviation by income on the matched same-job design and show that, once decomposed, it is two-signed: the bottom over-perceives its transitory risk even as, like every group, it under-perceives total risk, so the cross-section crosses its own benchmark rather than lying uniformly below it. Second, I document the within-person dynamics of perceived risk, its persistence and its response to realised shocks, where his evidence is cross-sectional. Third, I estimate an explicit updating rule, where he remains agnostic about belief formation by design. Fourth, I give the gradient a mechanism in how pay is delivered, measuring its primitive, the labelled share of transitory pay, from the itemised pay components of the same records, where he attributes the level gap to unobserved heterogeneity or overconfidence. The incidence differs accordingly: his beliefs move aggregate wealth and its dispersion, while the wedge here concentrates at the top, because the saving-relevant distortion, the under-perceived permanent risk, is itself graded by income and largest there.

The paper connects to several further strands. On subjective beliefs about income, Rozsypal and Schlafmann (2023) document an income-graded bias in expected income *levels*, where I study the second moment; Ben-David et al. (2018) show in the same survey that own-income uncertainty falls with income and education, and Koşar and van der Klaauw (2025) characterise the dynamics of the SCE's earnings expectations, to which this paper adds the same-job conditioning, the matched realised process, and the estimated rule. In linked Danish survey and register data, Caplin et al. (2026) likewise find survey-reported earnings risk to be several times lower than its indirectly estimated counterpart, and show that most of subjective earnings risk is attributed to job transitions, and that the risk conditional on staying in the same job is small and symmetric. Their instrument elicits earnings expectations conditional on each possible job transition, whereas the question here conditions on no job change, so the two measurements are complementary: they measure the risk that comes with changing jobs, while this paper measures the risk that remains within the job. Koşar and Melcangi (2025) study the margin the model prices, the response of the marginal propensity to consume to subjective earnings uncertainty. The mechanism speaks to the "insurance versus information" question in the partial-insurance literature (Blundell et al., 2008): Pistaferri (2001) and Kaufmann and Pistaferri (2009) use subjective expectations to separate anticipated from unanticipated shocks and show that superior information can rationalise part of the perceived-realised gap, which is the labelling channel here, made observable and graded by income.

The updating rule builds on Pischke (1995), who casts inference about permanent income as a signal-extraction problem, and the under-reaction it generates is the household counterpart of the information rigidity of Coibion and Gorodnichenko (2015), running opposite to the diagnostic over-reaction of Bordalo et al. (2018) that Sanchez and Song (2025) carry into an incomplete-markets model, so the sign of the news response separates the two accounts; experience-based learning (Malmendier and Nagel, 2011) instead grades the gain by age, where mine rises with income at fixed demographics.

Finally, the mechanism rests on documented features of how pay is delivered across the distribution, performance pay concentrated at the top (Lemieux et al., 2009), high-income shocks predominantly transitory (Guvenen et al., 2021), and volatile, fragmented pay at the bottom (Hardy and

Ziliak, 2014; Morduch and Schneider, 2017).

2 Data and measurement

The paper needs two objects that can be set against each other: what a worker believes about her wage risk, and the risk her wage actually carries. I measure the first from the New York Fed's Survey of Consumer Expectations (SCE) and the second from the Survey of Income and Program Participation (SIPP), both monthly within-respondent panels, the SIPP running over reference years 2013–2023 and the SCE over 2013–2025. I hold the two to the same conditioning, a worker who stays in the same job with the same employer for at least six consecutive months, so that any gap between them is a gap in beliefs and not in what is being measured, and I apply survey weights throughout. Respondents are sorted into three groups by median annual household income, the bottom below \$50k, the middle between \$50k and \$100k, and the top at or above \$100k.[1]

The perceived object comes from the SCE density question, which asks each respondent for the distribution of her own earnings growth over the next twelve months *"on the same job, same hours, same employer"*, reported as a ten-bin histogram in the density-forecast tradition of Engelberg et al. (2009). This conditioning is what makes the answer usable. By fixing the job, the hours, and the employer, it strips out the predictable path of pay and the part a worker varies herself, so the elicited object is the conditional variance of the stochastic component of the wage change rather than of income, the quantity Section 4 sets against the realised process. Two consequences follow. The measure is free of the voluntary labour-supply variation that confounds risk estimates built on total earnings, and it excludes the risk of losing or changing the job, the larger part of income risk (Caplin et al., 2026), which the model reintroduces separately through its employment block (Section 9). From each reported histogram I take the bin-midpoint variance as the perceived variance $\bar{\sigma}^2_{\text{perc}}$, exclude respondents who place all their mass in a single bin, trim at 1%/99%, and drop each respondent's reports from the first wave in which she records a job change.

The realised object comes from the SIPP, which I use in preference to the Panel Study of Income Dynamics because it carries a unique employer identifier and records earnings every month, so a worker can be held to the same job and employer at the frequency the SCE question asks about, where the biennial PSID cannot. Per-month earnings are regular wage and salary together with bonuses, commissions, tips, and overtime, the amounts actually paid, and the self-employed are excluded. I drop the January seam, where the reported cross-wave change bunches spuriously, and trim squared changes at 1%/99%.

I residualise both objects on the standard demographic controls before any moment is computed: age and its square, sex, education, state, and year-month fixed effects. This leaves the variation a worker's position in the income distribution explains, net of what her demographics and location predict. The SCE residualises the perceived variance $\bar{\sigma}^2_{\text{perc}}$ and the SIPP the log wage; year-month effects absorb aggregate shocks common to all groups. The SCE alone carries a survey-tenure term that absorbs the panel conditioning of repeated subjective reports (Kim and Binder, 2023), absent from realised pay.

[1] The cutoffs are nominal and fixed across the sample. The SCE skews toward higher incomes, so group sizes are uneven (estimation sample: 1,531 / 7,501 / 11,331 observations).

3 Perceived risk by income group

Levels

Perceived variance is steeply graded by income, with the bottom of the distribution far above a middle and top that sit together. The weighted group means are 20.6 at the bottom, 11.3 in the middle, and 11.4 at the top, so a worker at the bottom perceives about 1.8 times the wage uncertainty of one at the top, and Figure 1 traces it over the sample. The gradient takes the form of a step. The middle and the top are statistically indistinguishable and track each other closely. The gradient itself has an antecedent in falling own-income uncertainty in the same survey (Ben-David et al., 2018), where my added matched-stayer conditioning and the realised counterpart sharpen it into a statement about wage risk.

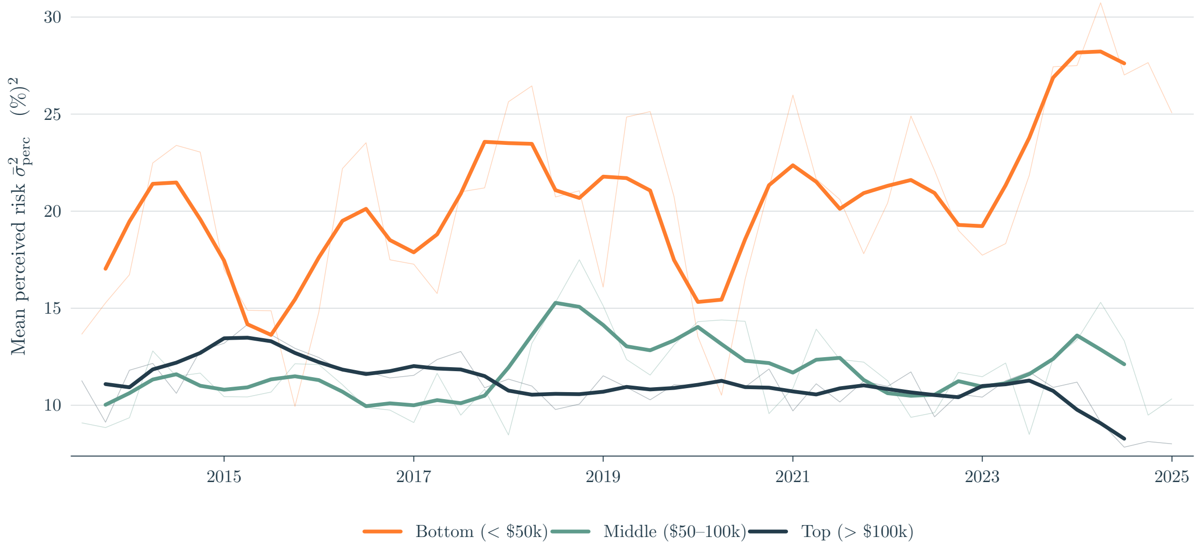

Figure 1: Group-mean perceived variance $\bar{\sigma}^2_{\text{perc}}$ by income group: quarterly means (thin) and their 4-quarter centred moving average (thick). Matched-stayer, trimmed and residualised SCE panel as in Section 2.

Two features of the sample challenge the level fact. The bottom series drifts upward over the sample, which may reflect the fixed nominal cutoffs or the greater cyclical exposure of low-income pay, but the estimated belief block is insensitive to it (Appendix C.2). And the bottom cut-off is set at $50k because the finer income brackets are too thin to support the estimation (Appendix E.1) and a finer split might have located more precisely where in the distribution the gradient forms.

The bottom group is also the least numerate, so a natural concern is that its higher reported risk is an artefact of how it reports. It is not: the gradient survives controls for numeracy and for inflation uncertainty (Appendix C.4), and the bottom uses more of the ten histogram bins than the top, not fewer, so its wider densities are not a coarser use of the histogram. If anything these controls understate the gradient, since the numeracy the bottom lacks is partly the ability to read its own fragmented pay, which later will be introduced as part of the mechanism; this is also why I keep occupation out of the baseline residualisation.

Persistence

Perceived risk is also more persistent within person at the bottom than at the top. I estimate a within-respondent AR(1) in the perceived variance $\bar{\sigma}^2_{\text{perc}}$, with persistence coefficient $\hat{\phi}_g$, by Blundell-Bond system GMM (Blundell and Bond, 1998). The short, wide SCE panel makes the within estimator severely Nickell-biased, so I instrument the lagged level with deeper lags to remove that bias, following Ma et al. (2020), who opts for this estimator for the same short-panel reason.

	Monthly		Quarterly	
Group	$\hat{\phi}_g$	SE	$\hat{\phi}_g$	SE
Bot	$+0.343^{***}$	0.046	$+0.632^{***}$	0.062
Mid	$+0.142^{***}$	0.026	$+0.594^{***}$	0.088
Top	$+0.144^{***}$	0.025	$+0.377^{***}$	0.076
Wald ($B = T$)	$p = 0.0001$		$p = 0.009$	

Table 1: Within-respondent AR(1) persistence $\hat{\phi}_g$ of perceived variance, Blundell-Bond system GMM. Instruments: $\bar{\sigma}^2_{\text{perc}}$ at lags 3–4 in differences plus the BB level moment, collapsed. Two-way clustered SEs (respondent, quarter). The same instrument set is used at both cadences.

The bottom is the most persistent of the three groups, and its equality with the top is rejected at both cadences. The gradient is also not inherited from the realised process, whose squared changes are if anything less persistent at the bottom (Appendix C.5). The quarterly estimates exceed the monthly ones for every group, a pattern the time aggregation of an AR(1) cannot produce, and classical noise in the monthly reports cannot fully produce it either, since the attenuation required would imply an implausibly persistent true process. I therefore read the two cadences as bracketing the true persistence and rest on what they agree on: the bottom is the most persistent and its gap to the top survives both cadences, while where the middle sits between them is not pinned. Taken with the level, the persistence leaves the standard heterogeneous-agent assumption that perceived risk equals realised risk with two gradients to reproduce: a calibration from realised wage data would have to match both the level and the persistence documented here.

4 The realised wage process

If the belief gradient reflected a gradient in actual risk, it would show up in the realised wage process, and it does not. Realised shocks nonetheless move beliefs, and together these two facts pin down what a belief rule must deliver.

Wage model

I use the standard permanent-transitory decomposition (Meghir and Pistaferri, 2004):

$$w_{i,t} = z_{i,t} + e_{i,t}, \qquad e_{i,t} = p_{i,t} + \theta_{i,t}, \qquad p_{i,t} = p_{i,t-1} + \psi_{i,t}.$$

Here $w_{i,t}$ is the residualised log wage, $z_{i,t}$ the predictable Mincerian path, and $e_{i,t}$ the stochastic component that is the focus of the rest of the paper. The permanent shock $\psi_{i,t}$ is iid with variance

σ^2_ψ and accumulates through the random walk $p_{i,t}$ (a merit raise or cost-of-living adjustment); the transitory shock $\theta_{i,t}$ is iid with variance σ^2_θ (a discretionary bonus or profit share).

Matching SCE to the model

The SCE question conditions on *same job, hours, employer*, which fixes $\Delta z = 0$, so the elicited variance is that of the stochastic component alone:

$$\bar{\sigma}^2_{\text{perc},i,t} \;\equiv\; \text{Var}_{i,t}(\Delta w_{i,t+1} \mid \Delta z = 0) \;=\; \text{Var}_{i,t}(\Delta e_{i,t+1}).$$

On SIPP I residualise (removing z) and difference to recover Δe directly. The recorded wage is the amount paid per usual hour, so it includes tips, overtime, and any gap between actual and usual hours: the recorded counterpart of the worker's paycheck. Both sides therefore measure $\text{Var}(\Delta e)$.

Identification of the wage primitives

The one-period change is $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$. Write $\bar{S}_m$ for the variance of this monthly change, $\mathbb{E}[(\Delta e)^2]$. Because ψ and θ are independent and iid, it has only two non-trivial moments per group:

$$\bar{S}_m = \sigma^2_\psi + 2\sigma^2_\theta, \qquad \text{Cov}(\Delta e_t, \Delta e_{t-1}) = -\sigma^2_\theta.$$

Group	σ^2_θ (transitory)	σ^2_ψ (permanent)	$\bar{S}_m$
Bot	2.52	11.89	16.92
Mid	4.42	11.60	20.44
Top	5.83	11.22	22.88

Table 2: SIPP permanent-transitory identification per income group, monthly cadence, in $(\%)^2$. Broad stayer panel, January seam dropped.

The lag-1 autocovariance identifies σ^2_θ and the remaining variance identifies σ^2_ψ, so the two wage primitives are exactly identified per group. Permanent variance is roughly flat across groups, while transitory variance rises mildly with income, plausibly a bonus channel at the top. The realised gradient therefore runs, if anything, *opposite* to the perceived one.

The FIRE benchmark

To compare the monthly process with the 12-month SCE horizon, I roll it forward twelve months under iid shocks (derivation in Appendix A):

$$\text{Var}(e_{t+12} - e_t) \;=\; 12\sigma^2_\psi + 2\sigma^2_\theta, \tag{1}$$

which is what a full-information rational-expectations (FIRE) agent who knows the parameters would report. The benchmark is the conventionally calibrated risk, near 148 $(\%)^2$ for every group, and perceived variance falls far below it, to 20.6, 11.3, and 11.4, between a seventh and a thirteenth of the benchmark (Table 6, Appendix A). That gap is also precisely the one Wang (2023) documents. Therefore, I place little weight on the absolute ratios; the argument rests on the cross-group contrast. The ratios overstate miscalibration because equation (1) multiplies permanent variance by twelve and transitory variance by two, so any contamination that loads onto σ^2_ψ is magnified twelvefold. SIPP earnings are self-reported, and reporting error is one such contaminant: its classical component differences into an MA(1) and is read as transitory, while any persistent component is indistinguishable from a true permanent shock and is multiplied by twelve (Bound and Krueger, 1991). Administrative payroll data place job-stayers' base-wage dispersion well below the benchmark for the same reason (Grigsby et al., 2021). This error is plausibly common across the groups, and I assume it cancels in the cross-group comparison, so it cannot sign the contrast.

Belief dynamics and realised shocks

The cross-section rules out the realised process as the *source* of the gradient, but the realised process can still drive belief *dynamics*: Figure 2 shows that pooled perceived risk co-moves with the absolute deviation of aggregate wage growth from its sample mean, consistent with surprises to wage growth moving perceived risk over time.[2]

Figure 2: SCE perceived risk and the absolute deviation of SIPP aggregate wage growth from its sample mean, both standardised, pooled across income groups, 2015–2022.

Together these facts impose two requirements on any belief rule. It must let perceived variance move, since the dynamics rule out constant beliefs; and it must let realised shocks drive that movement. Whether the response to a largely common signal differs by group is then an estimation question, taken up within person in Section 6.

[2]Absolute deviations, since any realised shock should raise perceived uncertainty; squared deviations give a very similar plot. The comovement motivates the shock regressor of Section 6; the within-person estimates there are the evidence that realised shocks move beliefs.

5 A belief-updating rule

I assume the agent cannot tell a permanent shock from a transitory one. Each period she sees a single squared pay change, which mixes the two, since $\mathbb{E}[(\Delta e)^2] = \sigma^2_\psi + 2\sigma^2_\theta$, and she must split that surprise, reading a share λ as news about her permanent earning power and the rest as transitory. The assumption concerns the residual a worker cannot label. Druedahl and Jørgensen (2020) infer from consumption responses that households track the persistent component of their income well, whereas the belief here is that a parallel second moment is formed from monthly pay changes. The mechanism of Section 7 will precisely microfound this assumption by granting the best-informed workers a cleaner separation of the labelled part of pay.

The rule carries two beliefs, a permanent belief $\hat{\sigma}^2_{\psi,t}$ and a transitory prior $\sigma^2_{\theta,\text{perc}}$, and the SCE gives one perceived-variance series per group, enough to pin down one but not both. I let the permanent belief evolve and hold the transitory at a static prior. The permanent belief is the welfare-relevant one, the object saving responds to, so it is the natural place for the dynamics. This is a normalisation, not a finding: the same series is equally consistent with a static permanent bias and a moving transitory belief, and any misallocation between the two loads onto λ. The transitory prior is its natural home because an unresolved reading error is recorded exactly as a transitory shock would be.

The permanent-variance belief follows an exponentially weighted moving average (EWMA) of past squared shocks, with the attribution share as a separate weight:

$$\hat{\sigma}^2_{\psi,t} \;=\; \underbrace{(1-\mu)}_{\text{persistence}}\;\hat{\sigma}^2_{\psi,t-1} \;+\; \underbrace{\mu}_{\text{update weight}}\;\underbrace{\lambda}_{\text{attribution share}}\;(\Delta e_{t-1})^2. \tag{2}$$

Here μ is the update speed, a higher μ moving the belief more on each period's news, and λ the share of each squared surprise read as informative about permanent earning power. Because $(1-\mu)$ and μ sum to one, the recursion is the integrated GARCH(1,1) in the squared surprise; it is equally the steady-state Kalman filter for an unknown permanent variance modelled as a random walk, with μ the Kalman gain (Appendix D.1 gives the filter and sets the gain by the signal-to-noise ratio). Given this information structure the agent applies the steady-state Kalman gain each period, the perpetual-learning rule for a drifting variance; what kind of departure from full information her constant attribution then represents, once legibility grades it, is taken up in Section 7.

Substituting the belief into the FIRE identity (1) gives the level the agent reports:

$$\bar{\sigma}^2_{\text{perc},t} \;=\; 12\,\hat{\sigma}^2_{\psi,t} \;+\; 2\sigma^2_{\theta,\text{perc}}. \tag{3}$$

Strict FIRE is the corner $\mu = 0$ with both beliefs at the truth, a constant reported variance that the dynamics of Section 3 rule out. Within the learning family ($\mu > 0$) FIRE survives only as a long-run average: taking unconditional expectations of (2) in a stationary state gives $\mathbb{E}[\hat{\sigma}^2_\psi] = \lambda\,\bar{S}_m$, so the belief is right on average exactly when $\lambda^* = \sigma^2_\psi/\bar{S}_m$, which Table 2 puts at 0.70, 0.57, and 0.49 across the three groups, with the transitory condition $\sigma^2_{\theta,\text{perc}} = \sigma^2_\theta$ alongside. These two conditions are what the estimates test.

6 Estimation

The belief recursion (2) is written in the unobserved $\hat{\sigma}^2_{\psi}$. Combining it with the level identity (3) eliminates the belief and leaves an AR(1) in the *observable* perceived variance:[3]

$$\bar{\sigma}^2_{\text{perc},i,t} = \underbrace{2\mu_g\, \sigma^2_{\theta,\text{perc},g}}_{\beta_{\text{cons}}} + \underbrace{(1-\mu_g)}_{\beta_{\text{lag}}}\, \bar{\sigma}^2_{\text{perc},i,t-1} + \underbrace{12\mu_g\lambda_g}_{\beta_{\text{shock}}}\,(\Delta e_{i,t-1})^2 + \alpha_i + u_{i,t}, \tag{4}$$

with α_i a person fixed effect. The three coefficients map one-to-one to the structural parameters: $\mu_g = 1 - \hat{\beta}_{\text{lag}}$, $\sigma^2_{\theta,\text{perc},g} = \hat{\beta}_{\text{cons}}/(2\mu_g)$, and $\lambda_g = \hat{\beta}_{\text{shock}}/(12\mu_g)$. Each is identified from a distinct feature of the data: β_{lag} from the within-respondent persistence, β_{cons} from the group level once the news loading is netted out, and β_{shock} from the response to the lagged realised shock.

The shock regressor

The rule calls for the agent's own $(\Delta e_{i,t-1})^2$. The SCE contains no individual wage histories, so I substitute the SIPP cell mean over workers in the same cell c = (group × Census division × month), following Guerreiro (2023); workers in the same local labour market share components of wage news, transmitted through coworkers and local media and salient for expectations (Kuchler and Zafar, 2019). The cell series is smoothed with a 3-month trailing MA.

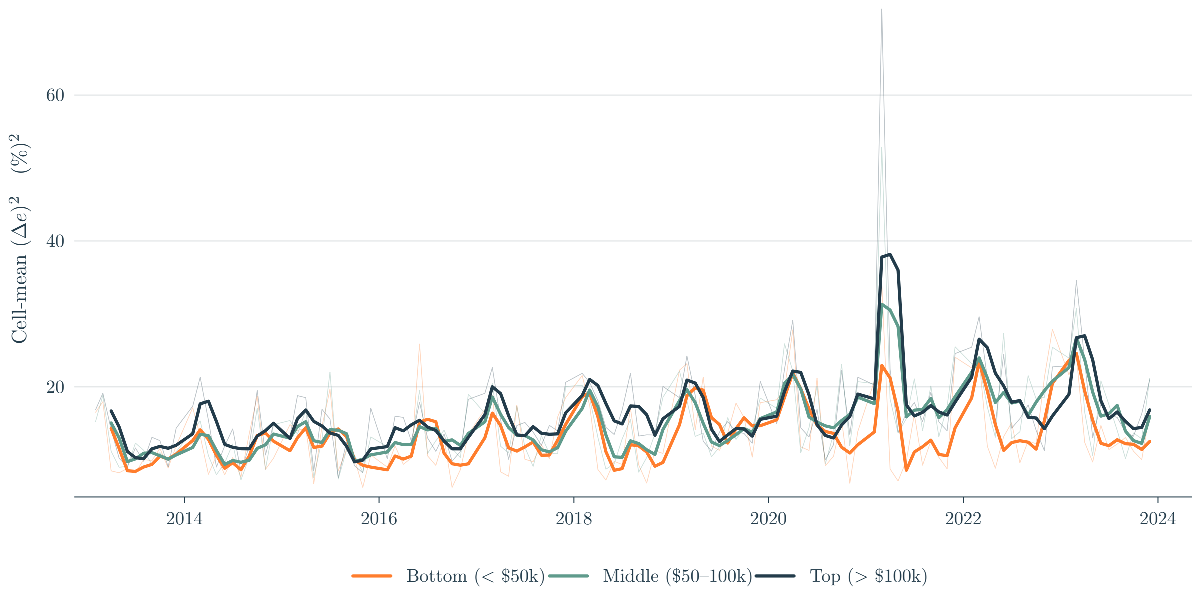

Figure 3: Cell-mean $(\Delta e_{i,t})^2$ by income group, national monthly, in $(\%)^2$. Thin lines: raw monthly cell means. Thick lines: the 3-month trailing MA used as the regressor in (4).

Figure 3 shows that the top has the largest shock amplitude, consistent with its higher σ^2_{θ} and with discretionary pay concentrating at the top (Guvenen et al., 2021; Halvorsen et al., 2024). Because the bottom's shocks are no larger than the top's, any stronger belief response at the bottom is a

[3] Solving (3) for the belief gives $\hat{\sigma}^2_{\psi,t} = (\bar{\sigma}^2_{\text{perc},t} - 2\sigma^2_{\theta,\text{perc}})/12$; substituting both belief terms into (2), multiplying by 12, collecting the intercept $(2\mu\sigma^2_{\theta,\text{perc}})$, and adding group subscripts, a person fixed effect α_i, and a shock $u_{i,t}$ gives (4).

property of its beliefs, the shocks themselves being if anything smaller.

Replacing the own shock with a cell mean is a measurement-error problem that attenuates β_{shock}, and hence $\hat{\lambda}$, toward zero. The attenuation works through finite-cell sampling noise: the cell mean tracks only the common component of a worker's shock, so its informativeness about her own shock depends on how large the common component is and on the cell size. The level of $\hat{\lambda}$ is therefore not interpretable on its own, and what I instead do is just read the cross-group pattern rather than its level.

I instrument with $\bar{\sigma}^2_{\text{perc}}$ at lags 3–4 and the shock at lags 1–3, together with BB level moments. The shock is treated as endogenous: the contemporaneous shock and the belief share a survey-timing and aggregate-news channel, and the Hansen J test rejects the contemporaneous moment set but does not reject lags 1–3.

Estimates by income group

Joint GMM on (4) by income group yields the estimates in Figure 4, set against the realised SIPP transitory variance. The full estimates, with standard errors, the Hansen J, and sample sizes, are in Table 7 (Appendix C).

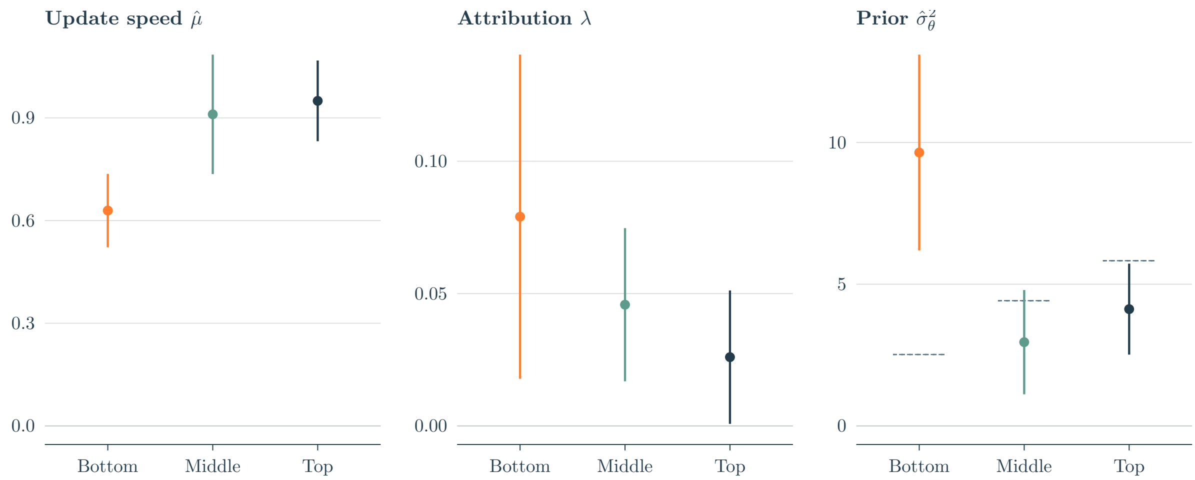

Figure 4: Headline estimates by income group: update speed μ_g, attribution λ_g, and transitory prior $\sigma^2_{\theta,\text{perc},g}$, with 95% confidence intervals. Grey dashes in the prior panel mark realised transitory variance σ^2_θ (SIPP).

Update speed. The bottom updates significantly more slowly than the middle or top, matching the persistence ordering of Table 1. The influence of a given wage shock on its belief therefore decays more slowly.

Attribution. Each $\hat{\lambda}_g$ is positive and significant, and the point estimates fall monotonically with income, the bottom's being three times the top's, although cross-group equality is not rejected. A common wage signal generates larger revisions of the permanent belief at the bottom, in point estimates if not yet in significance. Part of the decline is itself rational: the full-information share λ^* falls from 0.70 to 0.49 across the groups (Section 5), a gradient of about 1.4, so it is the excess of the estimated three-to-one slope over that benchmark that is belief content.

Transitory prior. Realised transitory variance rises with income; the perceived prior does the opposite. The bottom's prior sits almost four times its realised value, while the middle and top lie near or below theirs. The bottom is the only group whose prior is materially detached from its realised data, and the gradient is significant. So I conclude that the inflated prior is the dominant driver of the *level* of the bottom's perceived variance, separate from the news channel.

Robustness

Adding an occupation $\times$ month fixed effect to the shock construction preserves the persistence and attribution gradients and sharpens the latter, while the prior gradient slightly weakens (Appendix C.1); I do not adopt it as the baseline, since low-income workers sort into structurally more uncertain occupations (Morduch and Schneider, 2017), so partialling out occupation removes part of the channel.

Varying the instrument lags, the trim, and the COVID handling one at a time, the persistence gradient survives every variant except the tighter within-group trim, where it retains only marginal significance, and the attribution gradient stays monotone throughout (Appendix C.2). The prior's level excess at the bottom survives every variant, 6.4 to 11.2 against a realised 2.52, but its cross-group significance survives neither the COVID exclusion nor the occupation control. The COVID years are realised wage variation that the SIPP benchmark sample shares, so I read them as identifying variation rather than contamination; the occupation control removes cross-occupation transitory dispersion that itself feeds the bottom's excess, so I read the loss as over-control rather than evidence against it.

The bottom's perceived-variance series also trends upward over the sample, which could in principle drive the estimates through the level moments; it does not. Removing a group-specific linear trend or calendar-year effects before estimation moves the bottom's gain from 0.63 only to 0.62 and 0.64, leaves every gradient's significance intact, and a difference-in-Hansen test clears the Blundell-Bond level moments for every group (Appendix C.2). Lastly, with the shock split by sign and the groups pooled for power, both components load positively, upside news carrying about twice the weight of downside. I read the asymmetry as suggestive only: the extra weight may reflect the size of the surprise rather than its sign, since expected wage growth was positive over the sample, and pooling confounds the groups' responses (Appendix C.3).

7 A pay-delivery mechanism

The estimates of Section 6 give three parameters for each income group, the update speed, the attribution, and the transitory prior, each fitted freely, so the gradients across groups stand documented but unconnected. This section traces them to one feature of how pay is delivered: how legibly a worker's pay separates into a permanent and a transitory part. Legibility governs two forces of opposite sign, and the SIPP itemises the components of a paycheck, so it can be measured directly, on the same stayer panel that supplies the realised process.

The legibility of pay, and the bottom's prior

What the mechanism must explain is the shape of the transitory prior: far above realised risk at the bottom, close to it higher up. At the bottom the prior sits at 9.65 against a realised transitory variance of 2.52, a gap of nearly four to one that no plausible units or reporting-error correction closes; at the middle and the top it lies near the realised value, 2.96 against 4.42 and 4.12 against 5.83, gaps small enough that I read little into them, given that the two surveys' levels compare only loosely (Section 4). A single force common to every group would move each prior in the same direction, so an excess this large at the bottom and absent higher up requires two forces of opposite sign, each acting in one direction only: recognising part of transitory pay and netting it out before the update can only lower the prior, while pay too fragmented to read cleanly can only raise it, since an unresolved reading is recorded as a transitory shock. I conjecture that which of the two dominates, and so on which side of realised risk a group's prior falls, is a question of how its pay is delivered.

The downward force: labelling. A salaried worker at the top receives her transitory pay already flagged: a bonus or a commission arrives on its own line, a furlough comes with a return date, and the whole of her pay is a single salary to read against. A worker at the bottom receives the same kind of pay scattered and unlabelled, as variable shifts, cash tips, and supplements layered on uncertain hours, and her downturns arrive ambiguous, a sector closing or gig work drying up rather than a furlough that will end. The first worker can tell which part of a pay change is transitory and net it out before she updates; the second cannot. Three features of delivery grade this legibility by income: Lemieux et al. (2009) document that performance pay arrives on separate line items and concentrates at the top, Hoynes et al. (2012) that high-income macroeconomic downturns arrive classifiably where low-income ones do not, and Hardy and Ziliak (2014) and Morduch and Schneider (2017) that high-income pay is a single salary where low-income pay is fragmented. Let $\rho_g \in [0, 1]$ be the share of transitory pay a worker can separate and net out. The separated share is removed before she updates, so only the unlabelled residual feeds the belief recursion (2), and her prior falls to the unlabelled transitory risk that survives netting, $(1 - \rho_g)\sigma_\theta^2$, the floor that netting alone can reach.

The upward force: reading noise. A worker who has separated what arrives labelled still has to read what is left, and at the bottom that remainder arrives fragmented, as variable shifts, cash tips, and irregular supplements layered on uncertain hours (Morduch and Schneider, 2017): she cannot read cleanly how large the change even is, let alone how much of it will last. Her reading of the netted pay therefore carries an error,

$$\tilde{y}_t \;=\; p_t + \theta_t^u + \nu_t, \qquad \nu_t \overset{\text{iid}}{\sim} (0,\, \omega_g^2), \tag{5}$$

with p_t her permanent component, θ_t^u the unlabelled transitory component with variance $(1 - \rho_g)\sigma_\theta^2$, and ν_t the reading error, which leads to a signal-extraction structure in the style of Pischke (1995).[4] An unresolved reading is recorded as risk, and the prior settles above its netting floor by

[4]Recorded SIPP earnings capture the amounts actually paid, so ν never enters the moments of Section 4. The differenced error $\Delta\nu_t$ has variance $2\omega_g^2$, first autocovariance $-\omega_g^2$, and none beyond, exactly the signature of a differenced transitory shock, which is why an unresolved reading is recorded as transitory risk and why the prior is its natural home (Section 5).

exactly the variance of the error,

$$\sigma^2_{\theta,\mathrm{perc},g} \;=\; (1-\rho_g)\,\sigma^2_\theta \;+\; \omega^2_g, \qquad \omega^2_g \geq 0. \tag{6}$$

Measuring the labelled share

The measurement. The SIPP records, for each worker and month, her total earnings and the dollar amount of each pay component, bonuses, commissions, tips, and overtime, on the same-job stayer panel of Section 2. What I want is how much of a worker's pay *movement* arrives in a form she can recognise, so I begin by isolating that movement. Within each job spell I subtract from her total pay its spell average, which strips out the fixed level, so her base wage and whatever is constant about the job, and leaves only the month-to-month deviations; I then divide each deviation by spell-mean earnings, so a swing enters as a fraction of her typical pay and is comparable across workers who earn very different sums. What remains is how her pay moves within the job, and I split it in two. The labelled part is the pay that arrives flagged on its own line, bonuses and commissions; the unlabelled part is everything else, tips because they arrive as scattered cash and overtime because a variable shift count is the illegible delivery the mechanism is about. The labelled share ρ_g is the labelled fraction of everything that moves within a spell, the person-weighted regression share of the labelled deviation in the total, its covariance with the total over the variance of the total (Appendix D.2).

The denominator is the total within-spell movement of pay, which includes the drift in a worker's earning power and any tenure raise as well as the transitory variation. Because she cannot net that drift out, ρ_g lies below the labelled share of σ^2_θ the mechanism requires, a conservative estimate that removing the within-spell trend raises to the band's upper edge. The regression puts the labelled share at 0.121 at the bottom, 0.178 in the middle, and 0.358 at the top, rising with income and holding across all six specifications of Appendix D.2; at these shares the netting floors $(1-\rho_g)\sigma^2_\theta$ are 2.21/3.63/3.74, and any excess the estimated transitory prior holds above them is the reading noise ω^2 of (6), 7.43 (%)^2 at the bottom and indistinguishable from zero higher up (Appendix D.3).

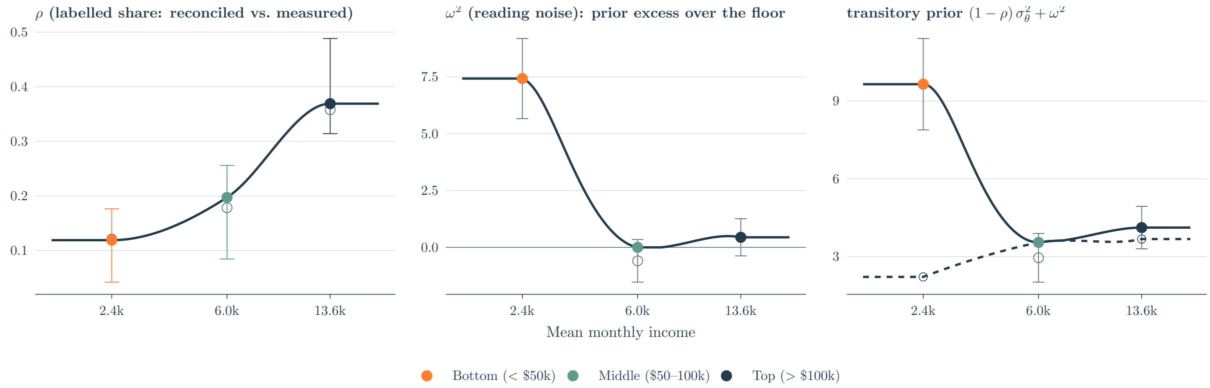

Figure 5: The labelled share ρ, the identified reading noise ω^2, and the transitory prior they compose (6), by income. Left: the reconciled share the model carries (filled points, with its income interpolant), against the main-specification measurement (open points) and whiskers spanning the measurement-specification band of Appendix D.2, from the imputation-dropped lower edge to the detrended upper edge; the belief moments select the carried share within the band (Section 9). Middle: the reading noise, identified as the prior's excess over its netting floor (the zero line), with the raw estimates' excess and their ±1 standard-error bands in grey; it is significant only at the bottom. Right: the carried prior (filled points, solid interpolant) as the sum of the netting floor $(1-\rho)\sigma_\theta^2$ at the carried share (dashed line, open ink points) and the reading noise, against the raw estimates and their ±1 standard-error bands in grey; the middle is carried on its floor, and the bottom sits the identified reading noise above it. In $(\%)^2$ except the dimensionless share.

From three groups to a profile. The model of Section 9 lets a worker's income drift continuously, so it needs the belief objects as functions of income rather than three points. I interpolate the carried share and the gain in levels and the prior in logs through the three anchors with monotone interpolants, holding each flat beyond the measured range and censoring the prior below at the local netting floor $(1-\rho(y))\,\sigma_\theta^2(y)$; the attribution and the permanent belief then follow from the closed forms at each income (Appendix E.1). The prior's dip at the middle survives, since the floors are non-monotone in the same way. Figure 5 draws the measured share, the identified reading noise, and the prior they compose against income, and Table 11 in the Appendix evaluates the mapping at representative incomes.

The update speed and the attribution

With the labelled share measured, the update speed and the attribution become predictions the estimates can be set against. Both fall with legibility through the same channel: labelling shrinks the part of a pay change the worker must still read, so a more legible worker faces a cleaner but smaller signal.

The update speed. How fast a worker updates depends on how clean a signal each pay change is. Labelling removes part of the transitory noise, shrinking the unlabelled residual variance $V_u = \sigma_\psi^2 + 2(1-\rho_g)\sigma_\theta^2$ relative to the whole change, so for a more legible worker each squared surprise is a sharper reading of her permanent risk and her steady-state Kalman gain is larger; the filter and the algebra are in Appendix D.1. The measured shares predict the gains' one pinned contrast correctly, the bottom updating slowest, but underpredict their spread, implying gains of 0.76/0.78/0.82 against the estimated 0.63/0.91/0.95 at the diagnostic drift (Appendix D.4). The model therefore uses the estimated gains rather than these filter-implied ones, since the filter only disciplines their ordering.

The attribution share. The attribution falls with legibility for a related reason. The rule has the agent book a single share λ of the whole squared pay change as permanent, but she has already set the labelled part aside as transitory and reacts only to the unlabelled remainder. The labelled part rides along in the change she sees without moving her belief, so the more of her pay arrives labelled, the larger the inert share of each change and the smaller the fraction of the whole she books as permanent. Under joint Gaussianity of ψ and θ this gives a closed form (Appendix B),

$$\lambda_g \;=\; \frac{\sigma_\psi^2\left(\sigma_\psi^2 + 2(1-\rho_g)\sigma_\theta^2\right)}{(\sigma_\psi^2 + 2\sigma_\theta^2)^2}, \tag{7}$$

which falls in ρ_g and equals the FIRE benchmark $\lambda^* = \sigma_\psi^2/\bar{S}_m$ at $\rho_g = 0$. At the measured shares it is 0.677/0.524/0.401, declining with income as the λ panel of Figure 4 does. The comparable moments are the cross-group ratios, since the cell-mean shock regressor attenuates the estimated level by a factor I treat as common across groups (Appendix D.3). Measured delivery predicts bottom-to-top and middle-to-top ratios of 1.69 and 1.31, against estimated ratios of 3.04 and 1.76 with log standard errors of 0.63 and 0.59, each prediction within one standard error of its estimate and about half of the estimated point gradient. Delivery therefore explains roughly half of how attribution declines with income, and the remainder is either an under-reaction beyond what labelling accounts for or sampling noise in ratios this wide.

8 What the estimated mechanism implies analytically

Section 7 measured a single pay-delivery primitive, the legibility ρ_g of a worker's pay, and carried the transitory prior alongside it as an estimated belief moment. In this section I work out what those two objects imply for behaviour, and I do so analytically, before the calibrated model of Section 9 attaches magnitudes to them.

The mechanism departs from full information in two ways: the worker nets out the labelled share of her transitory pay and stays unbiased about it, which therefore is genuine information updating; and also, she holds her attribution fixed as netting cleans her signal, which constitutes an under-reaction because the cleaner signal warrants a larger attribution to permanent risk than she applies, leaving her perceived permanent variance below the truth. Only the under-reaction creates a wedge in savings behaviour, and this section will analyse the extent.

I proceed in three steps. I first write the perceived second moments in closed form, isolate the one piece of them that drives precautionary saving, and collect the comparative statics in legibility. I then define the gap between perceived and true risk that organises the rest of the paper, and show that, since the error is based within the variance, it alters behaviour through prudence, where a level error would act through the intertemporal-substitution motive. I finally let the belief carry its own dynamics, so that a shock to the level of income and a shock to its risk leave wedges of very different durations.

The permanent belief in closed form

Averaging the constant-gain recursion (2) over a stationary state cancels the persistence terms and leaves $\mathbb{E}[\hat{\sigma}^2_\psi] = \lambda\,\bar{S}_m$, the attribution times the variance of the surprise she updates on. To rest at the truth she would need $\lambda^* = \sigma^2_\psi/\bar{S}_m$, the genuine permanent share of the full monthly change. Netting changes that surprise: setting the labelled transitory pay aside leaves a residual of smaller variance $V_u = \sigma^2_\psi + 2(1-\rho_g)\sigma^2_\theta < \bar{S}_m$ whose permanent part is still σ^2_ψ, so the residual is a cleaner signal, and a Bayesian agent would raise her attribution to $\sigma^2_\psi/V_u > \lambda^*$ and stay unbiased. The constant-gain agent I assume in Section 5 does not, since she cannot cleanly separate the permanent and transitory shocks, and so holds her attribution at the benchmark λ^* as netting cleans the signal; her belief settles at

$$\mathbb{E}[\hat{\sigma}^2_\psi] \;=\; \lambda^*\,V_u \;=\; \sigma^2_\psi\,\frac{V_u}{\bar{S}_m}. \tag{8}$$

The discount $V_u/\bar{S}_m \le 1$ is the under-reaction, the weight she applies as a fraction of the weight the cleaner signal warrants. The weight is exactly one when nothing is labelled ($\rho_g = 0$, so $V_u = \bar{S}_m$) and falls as she nets more, so the more legible her pay the further her perceived permanent risk sits below the truth.

An alternative reading is that the same belief reads equally as an attenuated attribution on the full change she observes: $\lambda = \sigma^2_\psi V_u/\bar{S}^2_m = \lambda^*(V_u/\bar{S}_m)$ of (7) sits below the benchmark λ^* and rests at the same belief, since $\lambda\,\bar{S}_m = \lambda^* V_u$. The labelled part rides in the change she reacts to without carrying news about permanent risk, attenuating her attribution exactly as classical measurement error attenuates a regression coefficient.[5]

Two features of (8) shape what follows. First, the prior does not enter, though its excess over realised transitory risk is largest at the bottom: the perceived prior enters both the agent's own reading and the recursion's gain by offsetting factors and cancels from the resting mean (Appendix D.1), so the reading error that inflates the prior survives in the transitory prior $\sigma^2_{\theta,\text{perc},g}$ and never in the resting permanent belief. Second, and as a corollary, the permanent belief that drives precautionary saving moves with legibility alone, so the saving distortion follows the legibility, and not the prior.

The perceived-risk wedge

The object that organises the rest of the section is the gap between the agent's perceived second moments and the true ones, summed across both shocks, which I call the perceived-risk wedge,

$$\Delta\sigma^2 \;=\; 12\big(\hat{\sigma}^2_\psi - \sigma^2_\psi\big) \;+\; 2\big(\sigma^2_{\theta,\text{perc}} - \sigma^2_\theta\big), \tag{9}$$

twelve times the permanent gap plus twice the transitory gap, the weights with which the two shocks enter the variance of the twelve-month income change. Its two pieces are signed differently,

[5] A third reading is the implemented one: the agent updates on the squared change of her own noisy netted reading (5), whose mean is $V_u + 2\omega^2_g = \sigma^2_\psi + V_x$ with $V_x = 2\sigma^2_{\theta,\text{perc},g}$, and the recursion's gain is the rescaled $\lambda^{\text{rule}} = \sigma^2_\psi V_u/[\bar{S}_m(\sigma^2_\psi + V_x)]$, chosen so that the resting mean is unchanged (Appendix D.1, Table 11): the reading error inflates her signal and the rescaling offsets it, so it never reaches the resting permanent belief.

and that difference generates the cross-section. The permanent gap $\hat{\sigma}^2_\psi - \sigma^2_\psi = \sigma^2_\psi[V_u/\bar{S}_m - 1] \le 0$ is one-signed, since legibility can only depress perceived permanent risk, while the transitory gap $\sigma^2_{\theta,\text{perc}} - \sigma^2_\theta = \omega^2_g - \rho_g\sigma^2_\theta$, by (6), is two-signed, increasing through the reading noise at the bottom and lowered by netting higher up. Because the two gaps pull in opposite directions, perceived total risk can sit on either side of the truth, and across groups it crosses over: the prior excess dominates at the bottom, where the wedge is positive, and legibility dominates higher up, where it turns negative. This is the prior crossing of Section 7, now stated as a property of $\Delta\sigma^2$ and drawn at the measured shares ρ_g in Figure 6, where the total crosses the benchmark only faintly and the sharp crossing comes from the transitory part alone.

Figure 6: Perceived twelve-month risk against the full-information benchmark, in closed form along the measured belief profile of Figure 5, split into its permanent and transitory parts. Each panel draws the benchmark (FIRE, solid) and the perceived belief (dashed); the distance between the two lines is the matching piece of the wedge (9). (a) the permanent part, $12\sigma^2_\psi$, with belief below throughout and the shortfall deepening as legibility discounts the permanent belief. (b) the transitory part, $2\sigma^2_\theta$, with belief above the benchmark at the bottom, where the reading noise inflates the prior, and below it higher up, where netting lowers it. (c) the total variance, $12\sigma^2_\psi + 2\sigma^2_\theta$ against $12\hat{\sigma}^2_\psi + 2\sigma^2_{\theta,\text{perc}}$, with belief above the benchmark at the bottom and crossing below from the middle on. Points mark the three income groups; in $(\%)^2$.

A gap in second moments reaches behaviour through the curvature of the value function. The continuation value $V(a,y)$ evaluated under the agent's perceived income density $\tilde{f}$ rather than the true f differs only through their spread, since the two share a mean, so a second-order expansion leaves the perceived-minus-true continuation as

$$\widetilde{\mathbb{E}}\, V(a,y') - \mathbb{E}\, V(a,y') \;\approx\; \tfrac{1}{2}\,\Delta\sigma^2\, V_{yy}(a,y), \tag{10}$$

the variance gap $\Delta\sigma^2$ times the income curvature V_{yy} of the value function, with $\widetilde{\mathbb{E}}$ and $\mathbb{E}$ the expectations under her perceived and the true density. As a second-order approximation it locates the channel and leaves its magnitude to the model of Section 9 where I conjecture that a perceived-risk gap reaches saving where the value function is most curved. The argument that follows rests on the prudence term of the Euler equation of which this expression is the compact complement.

The comparative statics of the next subsection make precise which part of the belief block carries the wedge, and the two subsections that follow trace it through prudence and through the belief's own dynamics.

Comparative statics in legibility

Every object in the belief block moves through the unlabelled residual variance V_u, which legibility lowers ($\partial V_u/\partial \rho_g = -2\sigma_\theta^2$). Differentiating the permanent belief (8), the gain through its observation noise $R_g = 2(\sigma_\psi^2)^2(V_u/\bar{S}_m)^2$ (Appendix D.1), and the attribution (7) through V_u gives the signs collected in Table 3, where $g_p \equiv \hat{\sigma}_\psi^2 - \sigma_\psi^2$ is the saving-relevant gap.

	Response to		
Belief object	ρ_g	ω_g^2	Margin it reaches
Permanent belief $\hat{\sigma}_\psi^2$	−	0	saving, through prudence
Saving-relevant gap $g_p \equiv \hat{\sigma}_\psi^2 - \sigma_\psi^2$	−	0	saving
Attribution λ_g	−	0	news response
Update gain μ_g	+	0	belief dynamics
Transitory prior $\sigma^2_{\theta,\text{perc},g}$	−	+	buffered, second order

Table 3: Comparative statics of the beliefs and the estimated rule in legibility ρ_g and the identified reading noise ω_g^2 of (6); entries are the signs of the partial derivatives. Legibility moves every object, while the reading noise enters only the transitory prior, one for one; the prior falls with legibility through the netting floor $(1-\rho_g)\sigma_\theta^2$. The zeros in the ω_g^2 column hold for the closed forms: in the recursion's own units the gain λ^{rule} absorbs ω_g^2 by construction (Appendix D.1).

Two things follow from the table. Legibility moves every object at once: it raises the update gain and lowers the attribution, so it governs how beliefs evolve and how they respond to news, and it is at the same time the only force holding perceived permanent risk below the truth. A single primitive thus sets the dynamics of beliefs and the saving distortion together. The reading noise enters only the transitory prior, one for one, and leaves the signal-to-noise ratio, the attribution, and the resting permanent belief untouched, since the recursion's own gain absorbs it by construction (Appendix D.1). Therefore, the saving-relevant gap is a matter of legibility alone: two groups that read their pay equally legibly carry the same gap however differently their priors are pinned, which is why the behavioural wedge widens with legibility along the income gradient and not with the prior.

Prudence and the incidence of the wedge

A belief about a variance moves saving through a different part of preferences than a belief about the level would. The saving choice is disciplined by the Euler equation, which weighs marginal utility today against its expected value next period, so a belief about future risk reaches saving only through the expected marginal utility $\widetilde{\mathbb{E}}[u'(c')]$, with $\widetilde{\mathbb{E}}$ and $\widetilde{\text{Var}}$ the expectation and variance under the agent's belief $\hat{\sigma}_\psi^2$. Expanding it around mean next-period consumption, $\widetilde{\mathbb{E}}[u'(c')] \approx u'(\widetilde{\mathbb{E}}c') + \frac{1}{2}u'''(\widetilde{\mathbb{E}}c')\,\widetilde{\text{Var}}(c')$, a belief about the mean would move the first term through u'', the curvature that fixes the elasticity of intertemporal substitution, while a belief about the variance touches only the second, through u'''. A perceived-variance error $\hat{\sigma}_\psi^2 - \sigma_\psi^2$ therefore acts on saving through prudence, just as a level error would act through the elasticity of intertemporal substitution. Carrying the consumption policy through,

$$\widetilde{\mathbb{E}}[u'(c')] - \mathbb{E}[u'(c')] \;\approx\; \tfrac{1}{2}\,(\hat{\sigma}^2_{\psi} - \sigma^2_{\psi})\,\mathbb{E}\big[u'''(c')\,c_y^2 + u''(c')\,c_{yy}\big], \tag{11}$$

with c_y and c_{yy} the first two income-derivatives of the consumption policy. Under the CRRA used here ($\gamma = 2$, so $u''' = \gamma(\gamma+1)c^{-\gamma-2} > 0$) the leading term is positive, so an agent who perceives less permanent risk than she faces under-saves for precaution, opening the wealth and marginal-propensity-to-consume wedge that Section 10 quantifies (Tables 12 and 13). The operative gap is the permanent one: a permanent surprise accumulates in the random walk and shifts the whole future consumption path, whereas a transitory belief error is buffered within the period and barely reaches c_y.

The incidence of this channel is gated by liquidity. Because perceived risk acts only through precautionary saving, the saving response to the permanent-variance belief, $\partial a'/\partial\hat{\sigma}^2_{\psi}$, is zero at the borrowing constraint $a' = \underline{a}$ and strictly positive only where the household holds slack, the opposite of a level belief error, i.e. a misperception of expected income rather than its risk, which reaches a constrained, high-MPC household one for one. In the hard-constraint limit the two incidences are disjoint: a variance error leaves the constrained household untouched and moves only the unconstrained one, a level error would do the reverse. In the cross-section, though, the ordering of the wedge follows the belief gap and not the constraint. The gap grades with legibility, small at the bottom, which labels little, and largest at the top; in the model of Section 10 the wealth wedge per unit of gap is nearly flat across groups, so liquidity reinforces the incidence at the margin without creating it. The bottom therefore holds the most distorted transitory prior of any group and still bears the smallest behavioural wedge: its distortion sits in the prior, buffered within the period, while its permanent belief is essentially unbiased.

Level shocks versus risk shocks

The wedge to this point is a stationary object, and how long a shock moves it depends on whether the shock is a one-off surprise or a lasting rise in risk. The permanent belief is a state driven by the squared surprise x_t, $\hat{\sigma}^2_{\psi,t} = (1-\mu)\hat{\sigma}^2_{\psi,t-1} + \mu\lambda(\rho)x_t$, so a single surprise raises it by $\mu\lambda(\rho)$ and then decays geometrically at rate $1-\mu$ ($\partial\hat{\sigma}^2_{\psi,t+h}/\partial x_t = \mu\lambda(\rho)(1-\mu)^h$), and the saving wedge inherits this path. A level shock, the kind a recession delivers, is a single such surprise, so it only traces this transient path: the belief jumps on impact and the constant gain discounts it away within a quarter, leaving no lasting wedge.

A risk shock, a persistent rise Δ in the true permanent variance, keeps the surprises large for as long as it lasts, but under the constant gain the agent books only the share $\lambda(\rho)$ of the increase as permanent, so her perceived permanent variance rises by $\lambda(\rho)\Delta$ and settles permanently short of the truth by the complement $\big(1-\lambda(\rho)\big)\Delta$. This standing shortfall is largest at the top, and it separates the way the attribution gradient does (Section 6): even a rational agent at the full-information share falls short by $(1-\lambda^*)\Delta$, largest where pay is most transitory, and legibility adds the excess $(\lambda^* - \lambda(\rho))\Delta$, again largest at the top. The top-concentration is thus shared with the rational benchmark, and only the excess is the mechanism's own; the bottom, attributing nearly its full rational share, barely under-perceives the rise, so the risk-shock wedge lands at the top for the same reason the steady-state one does. Section 9 embeds the measured beliefs in the quantitative model, and Section 10 takes both shocks to it.

9 A heterogeneous-agent model

To trace what those perceptions imply for behaviour, I embed the estimated belief structure in a closed general-equilibrium incomplete-market economy and compare each income group under its estimated beliefs with an otherwise identical full-information benchmark. The gap between the two regimes, in precautionary wealth and the marginal propensity to consume, is the belief wedge.

Setup: a general-equilibrium incomplete-market economy

The economy is a monthly overlapping-generations life-cycle buffer-stock model following Wang (2023), closed with a single-asset technology. A worker enters at age 25, works for forty years, retires at 65 onto a pension replacing sixty percent of her permanent income after a one-fifth step-down in the permanent wage at the boundary, and dies with certainty at 85, facing a small constant mortality risk before then; each death is matched by a new entrant drawn across the income distribution, so the cross-section is stationary through the finiteness of life rather than through the unit-root death-and-replacement device an infinite-horizon permanent random walk requires. The discount factor is a single value common across agents, calibrated to average liquid wealth as the calibration below takes up, so the cross-section of wealth and of marginal propensities to consume is an outcome of income and employment risk rather than of preference heterogeneity. Preferences are time-separable, constant relative risk aversion $\gamma = 2$ with a homothetic warm-glow bequest in terminal assets; retirement, its wage step-down, and the balanced-budget financing all follow Wang (2023).

Income follows the permanent-transitory process of Section 4 scaled by a deterministic age-wage profile, $y_{i,t} = \exp(p_{i,t})\, G_\tau\, \xi_{i,t}$, with $p_{i,t} = p_{i,t-1} + \psi_{i,t}$ the permanent random walk, G_τ a hump-shaped age profile, and $\xi_{i,t}$ the transitory state, the variances taken from Table 2. When employed the worker draws $\xi = \exp(\theta_{i,t})$; when unemployed she receives a benefit replacing half her permanent income; in retirement she receives the pension. The permanent innovation arrives while she is employed, consistent with the same-job stayer panel its variance is measured on, and it is the one she learns about in Section 5: accumulated over a career the innovations carry her across the income distribution, so a run of positive permanent raises lifts a worker out of the bottom group without any change of job, the deterministic profile G_τ adding a predictable component to the same drift; a current-income tercile therefore mixes types, a distinction the results return to. This is mobility in the sense the same-job perceived-risk question allows, the gradual drift of earning power rather than job-switching, consistent with the within-job conditioning of the elicitation. The measured labelled share, the update gain, and the perceived transitory prior are functions of current income, $\rho(y)$, $\mu(y)$ and $\sigma^2_{\theta,\text{perc}}(y)$, interpolated through the tercile anchors of Sections 6 and 7, and a worker slides along them as her income drifts. Following Carroll (2006) I normalise by permanent income: the state is cash-on-hand per unit of permanent income, the per-period draw is the transitory $\exp(\theta)$, and permanent shocks enter through the growth factor $G_\tau \exp(\psi)$. The problem is scale-invariant, so I report each group's mean monthly SCE income (\$2,415 / \$6,014 / \$13,604) only to fix magnitudes.[6]

Two regimes. I solve the model twice. Under the *belief* regime the agent perceives wage risk through the estimated rule and her belief evolves; under *FIRE* she knows the true variances. Everything else is shared, so any gap is the belief wedge. FIRE here means correct wage-risk beliefs; job-loss risk is common to both regimes and nets out.

Unemployment. The SCE measures *same-job* risk, so the budget adds a two-state employment block whose rates I take from the SCE itself, parallel to the wage-risk measurement and following Wang (2023): each respondent's perceived probability of losing her job within twelve months and of finding one within three months were she to lose it, survey-weighted by income tercile over employed respondents and converted to monthly Poisson rates.[7] This gives monthly separation 0.015/0.011/0.011 and job-finding 0.23/0.27/0.29 from bottom to top, a steady-state unemployment rate of 6.2/4.0/3.6 percent, higher and more persistent at the bottom, with replacement income half of permanent. The rates sit a little below the realised aggregate, a Shimer (2005) benchmark near seven percent, the known gap between perceived and realised job-loss risk; what the model uses is the perceived gradient, common to both belief regimes and perceived correctly, so it shapes the wealth distribution, the hand-to-mouth share, and the constraint incidence through which the wedge acts, in the manner Wang (2023) emphasises, but is not itself a source of the wedge. Beliefs update only while employed and freeze during unemployment.

Technology and market clearing. I close the economy with a constant-returns technology, $Y = ZK^{\alpha}N^{1-\alpha}$, capital share $\alpha = 0.36$ and monthly depreciation from an annual 2.5 percent, the single asset carrying both the household buffer and the return. The unemployment benefit and the pension are financed by flat balanced-budget taxes on labour income, so disposable income is net of a payroll and a social-security rate. The capital market clears against household asset supply, and I pin the productivity normaliser Z once from the FIRE economy at a target rate of three percent, holding it fixed across regimes so the FIRE-versus-belief rate gap is the general-equilibrium wedge.

The household problem

$$V_\tau(m,p,s,\hat{\sigma}^2_\psi) \;=\; \max_{0\le c\le m} \Big\{u(c)+\beta\,(1-D)\,\widetilde{\mathbb{E}}\big[V_{\tau+1}(m',p',s',\hat{\sigma}^2_{\psi'})\big]\Big\}, \qquad m' = \frac{R}{\Gamma'}\,(m-c)+\xi', \quad \Gamma' = G_\tau \exp(\psi'),$$

with normalised cash-on-hand m, age τ, income node p, employment status s, monthly mortality D, and $\widetilde{\mathbb{E}}$ the expectation under the agent's *perceived* process; the terminal age carries the warm-glow bequest in place of a continuation value. The budget uses *realised* income while expectations are taken under the perceived process, so beliefs act only through the saving motive: a higher $\hat{\sigma}^2_\psi$ raises perceived risk and hence precautionary saving through prudence ($u''' > 0$, Section 8). The agent observes realised income each period but cannot decompose a given month's change into

[6] Wage variances are stated in $(\%)^2$; in the income process they enter in fractional units (divided by 10^4) before exponentiating, and the perceived-variance state is carried back in $(\%)^2$.

[7] For each employed respondent I take the SCE's reported percent chance of losing the main job over the next twelve months and, conditional on losing it, of finding a new one within three months, average each within income tercile with survey weights, and convert a horizon-H probability P to a monthly hazard $-\log(1-P)/H$ (with $H = 12$ for separation and $H = 3$ for finding), the conversion Wang (2023) applies to the same SCE questions. The tercile-mean twelve-month job-loss probabilities are 16.9/12.7/12.2 percent and the three-month finding probabilities 50.2/56.0/57.6 percent.

permanent and transitory parts, the signal-extraction problem of Section 7; the budget is written in permanent-income-normalised form using her running estimate of permanent income, so the normalisation is an approximation, not a claim that she observes the permanent component exactly, and beliefs distort the perceived *variance* of future shocks, not the level of current income. The belief follows the constant-gain rule of Section 5, $\hat{\sigma}^2_{\psi,t} = (1-\mu(y))\hat{\sigma}^2_{\psi,t-1} + \mu(y)\,\lambda^{\text{rule}}(y)\,(\Delta\tilde{y}_t)^2$, on the agent's own netted reading $\tilde{y}$ of (5), where the gain is not free but the closed-form attribution (7) at the measured labelled share $\rho(y)$ rescaled to the reading's variance (λ^{rule}, Appendix D.1, Table 11), while the update speed $\mu(y)$ and the perceived transitory prior are carried as the estimated moments themselves (Section 7); it settles around $\lambda^{\text{rule}}(y)\,\mathbb{E}[(\Delta\tilde{y})^2] = \sigma^2_{\psi}\,V_u/\bar{S}_m$, the labelling discount holding it below the realised permanent variance wherever pay is partly labelled, mildly at the bottom and most at the top. The steady-state comparison below uses the resting value of this rule, which adds no state dimension; the dynamic state is carried explicitly for the risk-shock response.

Stationary equilibrium. For each regime a stationary equilibrium is a family of age-indexed value functions, policies $c_\tau(m, p, s, \hat{\sigma}^2_{\psi})$, and an invariant distribution Φ such that the policies solve the household problem, Φ is invariant under them, the capital market clears at the equilibrium interest rate r, and the government budget balances, the invariance running over the income process including the permanent drift across groups, the life-cycle demographics of entry, retirement, mortality and replacement, the employment transitions, and the belief law of motion (degenerate at the truth under FIRE). I read moments of Φ off a simulated panel of $N = 30{,}000$ households over 1,200 months (720 burn-in), with the belief and FIRE branches driven by a single common shock path, so each worker sits in the same income tercile under both regimes and the belief-minus-FIRE gap is a within-worker difference free of sampling noise.

Calibration

The calibration sets a single discount factor, common across agents and calibrated to average liquid wealth, and lets the belief block vary with income through the labelled share ρ and the carried estimates of Section 6; the share is the block's one pinned parameter, set jointly by its measurement band and the belief moments below, and nothing else in the block is fitted. The model is block-recursive: beliefs evolve on the realised income process and never on assets or consumption, so the belief block is measured and estimated independently of preferences. Table 4 collects the calibration.

Parameter	Symbol	Value (Bot / Mid / Top)	Source or target
A. Preferences and assets (common)			
Risk aversion	γ	2	Assigned (standard)
Annual interest rate	r	2%	Fixed for PE cross-section; clears near 3% in GE (Section 10)
Borrowing limit	$\underline{a}$	0	Assigned (standard)
Discount factor	β	0.917	Calibrated to mean liquid wealth ($\approx$ 1.8 mo)
B. Employment block (heterogeneous by income; common across regimes)			
Separation rate	s_g	0.015/0.011/0.011/mo	SCE perceived, monthly Poisson
Job-finding rate	f_g	0.23/0.27/0.29/mo	SCE perceived, monthly Poisson
Replacement rate	b	0.5	Standard convention
Pension replacement	ς	0.6	Standard convention
C. Realised income process by group, (%)^2			
Permanent variance	σ^2_ψ	11.9/11.6/11.2	SIPP (Table 2)
Transitory variance	σ^2_θ	2.5/4.4/5.8	SIPP (Table 2)
Mean monthly income	–	\$2,415/6,014/13,604	SCE (scale anchor only)
D. Beliefs by group: the labelled share pinned within its measured band, the rest implied or carried			
Labelled share	ρ_g	0.119/0.197/0.369	*Primitive*
Update gain	μ_g	0.629/0.910/0.950	Estimated GMM moment, carried (Section 6)
Attribution	λ_g	0.677/0.519/0.398	Implied from ρ_g, eq. (7) (App B)
Transitory prior	$\sigma^2_{\theta,\text{perc},g}$	9.65/3.55/4.12	GMM estimate

Table 4: Model calibration. Panel A: a single discount factor common across agents, calibrated so the FIRE economy matches average liquid wealth ($\approx$ 1.8 months); B: the employment block, perceived job-separation and job-finding from the SCE (Section 9), heterogeneous by income but common to both regimes; C: the realised SIPP process; D: the belief regime, where the labelled share ρ_g is pinned jointly, the measured SIPP band supplying the level and the attribution ratios and prior floors selecting within it (a within-block consistency diagnostic $J = 1.57$; see text); the attribution follows from it in closed form, the gain is carried as the estimated GMM moment, and the prior is carried as the estimate where it clears its netting floor and at the floor where it does not (the middle); under FIRE beliefs are held at the truth. The measured share carries a specification band, from an imputation-dropped lower edge to a detrended upper edge (Appendix D.2), which Section 10 carries through to the wedge.

Preferences, a single discount factor to average liquid wealth. Preferences are CRRA ($\gamma = 2$, $r = 2$ percent annual). I calibrate a single discount factor, common across agents, so that the FIRE economy matches the average liquid wealth in the data, about 1.8 months of permanent income across the three groups, which gives an annual β of 0.917.[8] This is the standard incomplete-markets discipline: it pins the *level* of wealth and of the marginal propensity while leaving the cross-section of both an outcome of risk. It is a middle course between two alternatives. Fixing the discount factor at a consensus value, as Wang (2023) does, leaves the level an outcome and the economy over-accumulates, liquid wealth running to tens of months and marginal propensities far below the data; calibrating an income-varying $\beta(y)$ to each group's wealth target instead builds the cross-sectional gradient into preferences. A single calibrated discount factor avoids both, hitting the average and letting the cross-section emerge as Section 10 reports it. Calibrating on the FIRE regime is a normalisation, not a substantive choice: under the measured beliefs average wealth falls short of the FIRE target by a few percent, so recalibrating β on the belief regime

[8] Targets from the SCE Household Finance module, the liquid-wealth concept of Kaplan et al. (2014): median liquid wealth about 0.5/2.0/3.0 months across the three groups, averaging near 1.8.

would move it only marginally and leave the wedge essentially unchanged. The discount factor sits just below the 0.95–0.96 of the life-cycle literature (Gourinchas and Parker, 2002), the price of holding a one-asset economy to 1.8 months of *liquid* wealth rather than to total net worth, while the twelve-month marginal propensities of 0.38–0.47 and hand-to-mouth shares of 41–43 percent lie near the empirical ranges of Kaplan et al. (2014). I solve by the endogenous grid method (Carroll, 2006) and simulate each regime on a common shock path.

Beliefs, pinned and carried. The labelled share is pinned by reconciliation: the SIPP measurement band of Section 7 supplies the level, and the belief moments select within it. The share triple minimises the standard-error-weighted distance to the measured shares themselves, with the specification band as their uncertainty, the estimated attribution ratios, and the admissibility of the carried priors against their netting floors $(1-\rho_g)\sigma_\theta^2$, which the mechanism does not let a prior undercut. The reconciled shares are 0.119/0.197/0.369, none more than a quarter of the band's standard error from its measured centre. The admissibility constraint moves one carried moment: the middle prior's estimate, 2.96, sits below its floor at every share in the band, so the fit carries it at the floor, 3.55, six tenths of a standard error from the estimate, while the bottom and top estimates clear their floors and are carried as estimated, 9.65 and 4.12. The fit is overidentified, five moments (the three banded shares and the two attribution ratios) against three parameters, and the reconciliation diagnostic reads $J = 1.57$ on two degrees of freedom ($p = 0.46$, read against a χ^2 heuristically since the share moments are weighted by a specification band rather than sampling error, and conservatively at that, the two attribution ratios sharing the top group's estimate so that weighting the pair by its full covariance lowers the statistic to 1.32): the pay records, the belief responses, and the prior floors are mutually consistent. Because the band is far wider than the shares' sampling error, they contribute little to J, which registers the consistency of the attribution ratios and the binding middle floor, not a test of the fitted share against its own measurement. The reconciliation is a consistency check, not a validation; the mechanism's tests are the measured-share scorecard of Appendix D.3 and the out-of-sample delivery variation the conclusion describes. The attribution the household block reads is the closed form (7) at the reconciled shares, 0.677/0.519/0.398; the gain is carried directly as the estimated GMM moment of Section 6, and the prior enters with its mild non-monotonicity intact, the floors being non-monotone the same way. The perceived permanent variance that drives saving then follows from the identity (8), a discount $V_u/\bar{S}_m$ of 0.96/0.91/0.81 on the realised permanent variance. Table 5 collects the pinned, implied, and carried objects side by side. The estimated attribution level, 0.079/0.046/0.026, is not comparable to the structural one, the cell-mean regressor attenuating it by an order of magnitude; the comparable moments are the cross-group ratios, which the reconciled shares predict to within one standard error, at about half the point gradient, the residual half being under-reaction beyond labelling or sampling noise (the scorecard of Appendix D.3).

	Pinned	Implied			Carried	
Group	ρ_g	$V_u/\bar{S}_m$	$\hat{\sigma}^2_\psi$	λ_g	μ_g	$\sigma^2_{\theta,\mathrm{perc},g}$
Bot	0.119	0.965	11.47	0.677	0.629	9.65
Mid	0.197	0.915	10.61	0.519	0.910	3.55
Top	0.369	0.812	9.11	0.398	0.950	4.12

Table 5: The belief block: pinned, implied, and carried. The labelled share ρ_g is pinned within its SIPP measurement band by the belief moments (this section); the discount $V_u/\bar{S}_m$, the resting permanent belief $\hat{\sigma}^2_\psi = \sigma^2_\psi V_u/\bar{S}_m$ of (8), and the attribution λ_g of (7) follow from it in closed form; the gain is the GMM estimate of Section 6 carried directly, and the prior the estimate carried where it clears its netting floor and at the floor where it does not (Mid); the bottom's carried prior is its floor plus the identified reading noise $\hat{\omega}^2 = 7.43$ of (6). Variances in $(\%)^2$.

Measurement and its band. The measured shares rise with income, as the mechanism and the performance-pay literature predict (Lemieux et al., 2009). The uncertainty that matters for them is specification rather than sampling, given the 2.25 million stayer months behind the point estimates: the main specification demeans each pay component within a job spell and takes the labelled share as the regression share of bonuses and commissions in total within-spell pay movement, and the alternatives bracket it, from a lower edge that drops months with imputed component amounts (0.042/0.085/0.314) to an upper edge that detrends rather than demeans each spell (0.176/0.256/0.488), the gradient surviving in every specification (Appendix D.2); Section 10 carries this band through to the wedge. A direct receipt-share counterpart in the same SIPP sample confirms the direction, placing the middle below the top on every labelled component (Appendix D.2).

What the measured beliefs imply. Table 12 (Appendix F) sets the stationary perceived variance against FIRE, with its permanent-belief and transitory-prior decomposition, in the same SIPP units, so the σ^2_ψ inflation that blocks an absolute comparison with the SCE (Section 4) cancels: what the model carries from the SCE is its cross-group ratios, its dynamics, and the prior gradient, never the absolute level the two surveys do not share. Perceived twelve-month risk falls with income, from 153 through 139 to 124 against a FIRE benchmark flat near 148, a ratio of 1.03/0.94/0.84: the bottom perceives about three percent more total risk than it faces, its inflated transitory prior outweighing a tercile-mean permanent discount of about four percent, while the middle and the top perceive six and sixteen percent less, the labelling discount deepening with income faster than the prior falls. The crossing sits in the components, as Section 8 derives: the permanent-variance belief that drives saving is discounted least at the bottom and most at the top, while the prior excess lives at the bottom alone. The current-income bottom tercile mixes in higher-income workers on a weak month, so a bottom-income *type*, which labels little ($\rho = 0.12$), carries a nearly unbiased permanent belief and little saving-relevant distortion, and the saving wedge the bottom *tercile* still shows is partly composition, which the workers-only panel of Table 13 isolates.

10 General equilibrium

The closed economy

Closing the economy lets the saving wedge move the interest rate. At the calibrated discount factor the FIRE economy clears at the three-percent target by construction, with the productivity normaliser Z pinned once from FIRE and held fixed across regimes (Section 9); the belief economy, under-perceiving permanent risk, supplies about five percent less aggregate capital and clears at a rate 0.21 percentage points higher, 3.2 against 3.0 percent, with the wealth Gini 0.004 higher (Table 14, Appendix F). This is the choice channel of Wang (2023) in general equilibrium: lower perceived risk lowers precautionary supply, raises the equilibrium return, and concentrates wealth, though under the measured shares the aggregate movement is modest, the wedge showing in the cross-section more than in the aggregates. The magnitude is the one-asset response, in which a single liquid asset carries both the buffer and the return; a credible quantification of how the wedge moves marginal propensities and the transmission of aggregate shocks needs a two-asset closure separating liquid from illiquid wealth.

The cross-sectional wedge

Figure 7 compares the belief regime with an otherwise identical FIRE benchmark at the calibrated interest rate; the underlying numbers are in Table 13 (Appendix F). With the discount factor calibrated to average liquid wealth, the FIRE economy holds 2.3/1.7/1.5 months of permanent income, with twelve-month MPCs of 0.47/0.41/0.38 and hand-to-mouth shares of 43/41/42 percent. The propensities sit in the data's range and the shares a little above it; mean wealth in months falls mildly with income where the median in the data rises, a miss that survives the workers-only split of Table 13. The direction is what a one-asset buffer-stock model produces: wealth in months is a normalised buffer, governed by the risk a group faces rather than by its income, and the risk a buffer defends against, persistent unemployment above all, falls with income here, so a flat risk gradient would level the profile rather than lift it. The rising liquid-wealth profile in the data is accordingly not precautionary saving by the rich but the life-cycle, illiquid, and bequest saving that a single asset routes through the same buffer, which income-varying patience would reproduce; the single common discount factor forgoes it to leave the cross-section an outcome of risk rather than of preferences. The object of interest is the belief wedge, computed with this calibration held fixed.

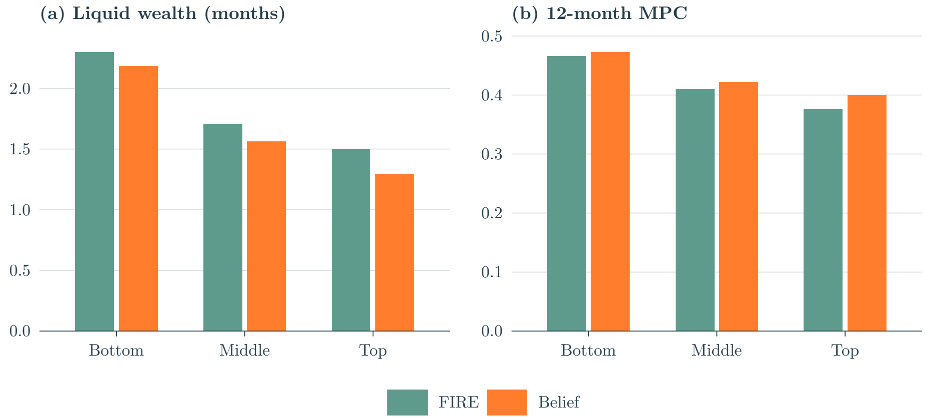

Figure 7: Mean liquid wealth (months of permanent income) and the 12-month MPC by income group, FIRE versus the belief regime (Table 13).

The wedge in wealth rises monotonically with income, the belief economy holding 5.1, 8.3, and 13.8 percent less than FIRE from bottom to top under the reconciled shares, and 7.0, 10.9, and 17.6 percent at the detrended upper edge of the measurement band (Appendix D.2), while the wedge in the marginal propensity to consume concentrates at the top, +0.7, +1.3, and +2.4 points, and +3.1 at the top at the detrended upper edge. Both profiles track the same object, the permanent-belief gap, smallest at the bottom and largest at the top, so the cross-section of the wedge is essentially the cross-section of the gap, and liquidity reinforces the incidence at the margin without creating it. A decomposition makes the same point exactly. Solving the model with the permanent discount alone, the transitory prior held at the truth, reproduces the full wedge on both margins, within a percent of it in wealth and exactly in the marginal propensity; the complementary regime, the carried prior with the permanent belief held at the truth, moves neither wealth nor the marginal propensity measurably. The prior is behaviourally inert in this economy; the discount is the wedge.

The bottom is least moved on either margin, +0.7 points on the MPC and 5.1 percent on wealth, much as Section 8 anticipates: a bottom-income type labels little ($\rho = 0.12$), so its permanent belief is discounted by under four percent and carries the least saving-relevant distortion. Composition does part of the rest. Just over half of the bottom current-income tercile is retirees, against roughly a quarter and a tenth above, and among its workers the bottom is in fact the least constrained of the three groups, with hand-to-mouth shares of 27/38/38 percent from bottom to top, so the smaller wedge is a fact about beliefs and composition, not about a binding constraint; restricting to workers and re-terciling among them, the wedge is −5.8/−9.7/−15.1 percent, the same shape mildly sharpened. The middle is moderately distorted on both margins (−8.3 percent on wealth, +1.3 points on the MPC): unconstrained and labelling more of its transitory pay, it under-saves. The top responds most on both, holding 13.8 percent less wealth and consuming over two points more of a windfall over the year, the group that labels its transitory pay away most completely, under-perceives the permanent risk that drives saving most, and acts on it. The labelled belief lies below the truth at all times, so the wedge is a steady-state object, not merely a response to shocks.

An upper bound. The measured calibration reads the belief estimates through the measured delivery of pay; a variant instead takes the estimated attribution responses at face value, anchoring the bottom at its full-information share and letting the estimated attribution ratios set the middle and the top, $\lambda = 0.702/0.409/0.231$, so the resting permanent belief falls to 11.89/8.36/5.29, discounts of 1.00/0.72/0.47 against the reconciled 0.96/0.91/0.81 (Appendix E.2). Its wedge is 8.5, 19.5, and 32.4 percent of wealth and +1.2, +3.1, and +6.7 points on the marginal propensity, the same shape at roughly twice the size. The variant shows what the estimated belief responses imply when delivery is asked to carry the whole of their gradient; the gap between it and the measured calibration is the part of the wedge that rides on netting beyond the measured pay components, or on sampling noise in attribution ratios whose standard errors admit either reading.

11 Dynamics

The belief is a state, so the wedge carries its own dynamics, and whether a shock leaves a lasting one turns on what the shock does to that state. With prices held fixed I trace two shocks through the belief channel alone, the dynamic-belief solution of Section 9 against an otherwise identical full-information economy driven by the same shock path, so the belief-minus-FIRE gap is a within-worker difference free of sampling noise. The first shock is to the level of income, the kind a recession delivers; the second is to income risk itself. The two leave wedges of very different durability, the distinction Section 8.5 draws analytically.

A level shock. A single three percent permanent fall in the common component of income, traced against each regime's own no-shock path (Figure 8). Consumption falls with the permanent drop in both regimes, and the belief regime's perceived risk jumps on impact, by twenty-five to fifty $(\%)^2$ as the squared surprise enters the signal, but the belief channel adds only a few hundredths of a percent to the consumption response and reverts within a quarter, and liquid wealth never separates. This is the transient path of Section 8: a single squared surprise raises the belief stock only briefly, the constant gain discounting it geometrically away, so the wedge lives in the chronic level of perceived risk and not in the reaction to any one shock. A recession, on its own, opens no belief wedge.

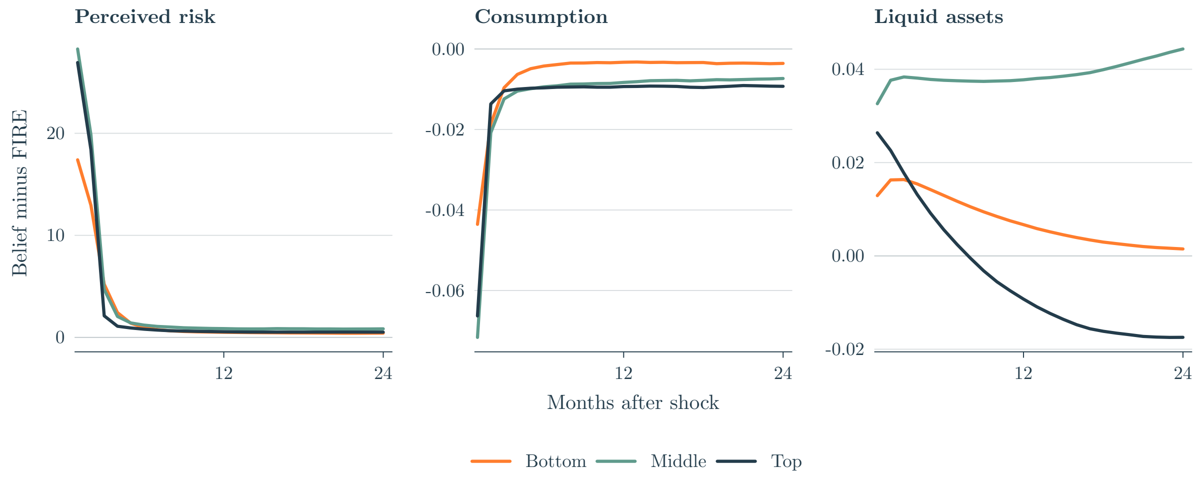

Figure 8: Response to a single 3 percent permanent (recession) income fall, belief regime minus FIRE, by income group, on independent scales. Perceived risk in $(\%)^2$; consumption and liquid assets in percent of the no-shock baseline. Paths average six simulated cohorts; the perceived-risk panel, which reads the belief-state mean, is shown as a three-month moving average.

A risk shock. A persistent forty percent rise in the true permanent variance, the second-moment shock that drives time-varying precaution (Figure 9). Full information recognises it at once and rebuilds its buffer, holding fifteen to nineteen percent more liquid wealth within two years; the belief regime books only the share λ^{rule} of the now-larger squared surprises as permanent, 0.37/0.57/0.47 across the groups on the agent's own netted reading (the implemented form of Section 8.5), so its perceived permanent risk rises far less and it barely adds to its buffer. It therefore accumulates 14.6/18.2/18.9 percent less additional liquid wealth than full information over the two years after the shock, each regime read against its own no-shock path, so this is the shock-response gap and not the total level gap, which adds the resting wedge on top, still widening at that horizon, and consumes about two points more of income on impact and a point and a half by two years. The perceived shortfall is largest at the bottom, whose reading noise damps its passthrough most, but the behavioural response concentrates at the top: the bottom under-perceives the rise most and yet sits too close to its constraint to act on it, while the top under-perceives less but is unconstrained enough to let the smaller distortion move its saving. This is the dynamic counterpart of the steady-state cross-section, the bottom holding the most distorted belief and the top bearing the wedge, and the shortfall decomposes as Section 8.5 sets out, into a constant-gain part a full-information agent on the same noisy signal would also carry and a legibility excess that is the mechanism's own, largest at the top. In general equilibrium such a shock would move the interest rate by shifting aggregate precautionary supply, with the top dominating the response.

Figure 9: Response to a persistent 40% rise in the true permanent wage variance, belief regime minus FIRE, by income group, on independent scales. Perceived risk is the gap in the rise in twelve-month variance, in $(\%)^2$; consumption and liquid assets are in percent of the no-shock baseline. Paths average six simulated cohorts; the perceived-risk panel, which reads the belief-state mean, is shown as a three-month moving average.

12 Conclusion

The paper opened by asking whether the risk heterogeneous-agent models assign their agents is the risk those agents perceive, and whether any gap between the two falls evenly across households or pulls them apart. It is not, and it pulls them apart. Conditional on staying in the same job, workers at the bottom of the income distribution report about 1.8 times the wage uncertainty of workers at the top, hold that belief more persistently, and revise it more strongly after a realised shock, against a realised process that shows no counterpart to any of the three. A single constant-gain rule organises the three gradients within person; the persistence and prior gradients are statistically significant, the attribution gradient monotone in point estimates.

The mechanism the paper attaches to the rule is measured, not conjectured: the legibility of pay, the share of transitory pay that arrives labelled and is netted out before the update, is measured directly from the itemised pay components of the same SIPP panel, rising from about an eighth of transitory pay variance at the bottom to over a third at the top, and the estimated belief moments then become predictions it is scored against. The measured netting floors place the middle and top priors within a standard error of where they sit, isolate the bottom's reading-noise residual at four standard errors from zero, and predict the attribution ratios within a standard error at about half the point gradient, a single primitive generating the persistence of beliefs, their response to news, and the saving distortion together; the share is the block's one fitted parameter, pinned inside its measured band. Embedded in a life-cycle economy under one calibrated discount factor, the measured beliefs open a precautionary wedge that rises with income and concentrates at the top, the group whose perceived permanent risk is discounted most; it is larger under the upper-edge and belief-anchored variants, and in one-asset general equilibrium it moves the capital stock and the interest rate only modestly, showing in the cross-section more than in the aggregates.

The natural test of the mechanism is out of sample, in variation in how pay is delivered rather than in how much it varies: performance-pay shares across occupations and reforms to payroll formats move legibility while holding risk fixed, and the rule says whose beliefs they should move. The test is sharper now that the primitive is observable: a payroll reform shifts a labelled share that can be measured before and after in the same records, so the comparison is between two measurements and a predicted belief response, not between a fitted parameter and a hoped-for proxy. A recession and a risk shock, traced through the dynamic belief state in Section 11, show the wedge to be a feature of the chronic level of perceived risk rather than of the reaction to any single shock; a two-asset closure in which the wedge's effect on marginal propensities can be credibly priced remains for future work.

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A 12-month FIRE variance

Rolling $e_t = p_t + \theta_t$ forward H months, the permanent piece accumulates one fresh innovation per month, $\mathrm{Var}(p_{t+H} - p_t) = H\sigma_\psi^2$, while the iid transitory piece contributes only its endpoints, $\mathrm{Var}(\theta_{t+H} - \theta_t) = 2\sigma_\theta^2$. The shocks are independent, so $\mathrm{Var}(e_{t+H} - e_t) = H\sigma_\psi^2 + 2\sigma_\theta^2$, which at $H = 12$ is (1).

Group	FIRE benchmark ($12\sigma_\psi^2 + 2\sigma_\theta^2$)	SCE observed $\bar{\sigma}^2_{\text{perc}}$	ratio
Bot	147.7	20.6	7.2×
Mid	148.0	11.3	13.1×
Top	146.3	11.4	12.8×

Table 6: FIRE-implied 12-month variance versus SCE-reported perceived variance, in $(\%)^2$.

B Closed form for λ_g under partial labelling

Split the transitory shock $\theta_t = \theta_t^l + \theta_t^u$ into independent pieces with variances $\rho_g\sigma_\theta^2$ and $(1-\rho_g)\sigma_\theta^2$. Then $\Delta e_t = u_t + l_t$, with unlabelled residual $u_t = \psi_t + \theta_t^u - \theta_{t-1}^u$ and labelled component $l_t = \theta_t^l - \theta_{t-1}^l$, so $V_u = \sigma_\psi^2 + 2(1-\rho_g)\sigma_\theta^2$, $V_l = 2\rho_g\sigma_\theta^2$, and $V_u + V_l = \sigma_\psi^2 + 2\sigma_\theta^2$.

The informative part of a pay change is u_t: were the worker to observe u_t^2 alone and re-optimise her gain for that cleaner signal, she would set it to σ_ψ^2/V_u and be unbiased, $\mathbb{E}[\hat{\sigma}_\psi^2] = \sigma_\psi^2$. That is the no-attenuation benchmark, not the attribution she carries. The rule of Section 5 updates on the full observed change $(u+l)^2$, so the attribution identified by (4) is the benchmark gain attenuated by the labelled part l that rides in that change, $\lambda_g = (\sigma_\psi^2/V_u)\,\mathrm{Cov}(u^2, (u+l)^2)/\mathrm{Var}((u+l)^2)$. Under joint Gaussianity with $u \perp l$, $\mathrm{Cov}(u^2, (u+l)^2) = 2V_u^2$ and $\mathrm{Var}((u+l)^2) = 2(V_u + V_l)^2$, so

$$\lambda_g = \frac{\sigma_\psi^2 V_u}{(V_u + V_l)^2} = \frac{\sigma_\psi^2\left(\sigma_\psi^2 + 2(1-\rho_g)\sigma_\theta^2\right)}{(\sigma_\psi^2 + 2\sigma_\theta^2)^2},$$

which is (7).

The closed form is one of two readings of the estimating equation, and the scorecard does not turn on the choice. Under the alternative, the agent nets first and updates on the residual u_t^2 while holding the no-labelling attribution $\lambda^* = \sigma_\psi^2/\bar{S}_m$ fixed; the coefficient on the full change in (4) is then the projection of her update on it, $\lambda^*\,(V_u/\bar{S}_m)^2 = \sigma_\psi^2 V_u^2/\bar{S}_m^3$, and the predicted bottom-to-top and middle-to-top ratios become 1.99 and 1.48 in place of 1.69 and 1.31. Both pairs lie within one standard error of the estimated ratios (Appendix D.3), so the ratio comparison accepts either form; the implemented model updates on the agent's own netted reading with the rescaled gain λ^{rule} (Appendix D.1), which shares its resting mean with both.

C Belief estimates and robustness

Table 7 reports the headline joint-GMM estimates of (4) plotted in Figure 4. The rest of this appendix checks their robustness: C.1 adds an occupation × time fixed effect to the shock construction, C.2 varies the instrument lags, the trim, and the COVID handling one at a time, C.3

reports the sign-split shock, C.4 conditions on numeracy and inflation uncertainty, C.5 checks the persistence of the realised volatility process, and C.6 checks the realised process itself, its higher-order moments and the stayer conditioning.

Group	σ^2_θ (SIPP truth)	$\sigma^2_{\theta,\text{perc},g}$ (GMM)	λ_g (GMM)	μ_g (GMM)	Hansen J p	N
Bot	2.52	**9.65** (1.76)	+**0.079** (0.031)	0.629 (0.055)	0.23	1,531
Mid	4.42	2.96 (0.94)	+0.046 (0.015)	0.910 (0.089)	0.23	7,501
Top	5.83	4.12 (0.82)	+0.026 (0.013)	0.950 (0.060)	0.23	11,331
Wald ($B = T$)	—	p = **0.004**	p = 0.12	p = **0.0001**	—	—

Table 7: Joint Blundell-Bond system GMM on (4), by income group (the estimates of Figure 4). Two-way clustered SEs (respondent, quarter) in parentheses. Instruments: $\bar{\sigma}^2_{\text{perc}}$ at lags 3–4, the shock at lags 1–3, plus BB level moments, collapsed. N counts observations; the respondent counts are 269/1,050/1,452. The Hansen J never rejects.

C.1 Occupation × time fixed effect

Adding an occupation × year-month fixed effect to the wage-shock construction (the interaction is the active term, since an additive occupation effect cancels in the within-spell difference) preserves the persistence and attribution gradients and sharpens the attribution; the prior gradient holds in direction but loses cross-group significance, so part of the bottom's inflated prior reflects cross-occupation transitory dispersion. The labelling channel behind λ is robust to occupation controls; the reading-noise channel behind the prior operates partly through cross-occupation variation.

	Canonical			Occupation × time FE		
Group	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}^2_{\theta,\text{perc},g}$	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}^2_{\theta,\text{perc},g}$
Bot	0.629 (0.055)	+**0.079** (0.031)	**9.65** (1.76)	0.649 (0.061)	+**0.088** (0.039)	**7.30** (2.66)
Mid	0.910 (0.089)	+0.046 (0.015)	2.96 (0.94)	0.923 (0.090)	+0.037 (0.010)	2.62 (0.98)
Top	0.950 (0.060)	+0.026 (0.013)	4.12 (0.82)	0.959 (0.060)	+0.017 (0.008)	4.19 (0.73)
Wald μ (T−B)		p = **0.0001**			p = **0.0003**	
Wald λ (B−T)		p = 0.116			p = 0.068	
Wald σ^2 (B−T)		p = **0.004**			p = 0.260	

Table 8: Canonical specification versus the same regression with an occupation × year-month fixed effect in the wage-shock construction. Standard errors in parentheses.

C.2 Lag depth, trim, and COVID exclusion

Each row of Table 9 varies one element of the canonical specification. The persistence and attribution gradients are robust across the lag variants, with the attribution Wald tighter than canonical in all four. The within-group trim preserves the qualitative pattern but loosens both Walds. Excluding COVID sharpens the attribution gradient and breaks the prior gradient's significance, so part of the bottom's elevated prior is identified from the COVID year. The Hansen J never rejects.

The bottom series drifts upward over the sample (+0.83 per year, t = 3.7), which strains the mean stationarity behind the Blundell-Bond level moments that identify the constant, and through it the prior. The prior does not rest on them: removing a linear calendar trend or calendar-year effects before estimation leaves it essentially unchanged (9.74 and 9.57 against 9.65), dropping the level instruments entirely leaves it at 9.22 (1.81) while costing precision only on the gain, and the split-sample estimates of 6.08 and 10.46 remain well above the realised 2.52. The same variants applied to the rest of the belief block change nothing: under a group-specific linear detrend or calendar-year effects the gains are 0.62/0.91/0.95 and 0.64/0.91/0.95 against the canonical 0.63/0.91/0.95, with the gain Wald below 0.0001 throughout and the prior Wald at most 0.008, and an AR(1) persistence coefficient estimated on the same panel behaves identically (bottom 0.41 canonical against 0.40 detrended, bottom-to-top Wald never above 0.0005). A difference-in-Hansen statistic isolating the level moments does not reject for any group (C = 1.20 and 1.63 on two degrees of freedom at the bottom and the top, p = 0.55 and 0.44; the middle's statistic is negative, no evidence against), while dropping the levels entirely preserves the point estimates but destroys the gain's precision: the level moments carry the identification and are not rejected by the test that isolates them.

	$\hat{\mu}_g$			$\hat{\lambda}_g$			$\hat{\sigma}^2_{\theta,\text{perc},g}$			Wald p (B vs T)		
Variant	B	M	T	B	M	T	B	M	T	μ	λ	σ^2
Canonical	0.629	0.910	0.950	+0.079	+0.046	+0.026	9.65	2.96	4.12	**0.0001**	0.116	**0.004**
Trim 2.5/97.5	0.824	0.846	0.891	+0.096	+0.055	+0.014	11.16	4.05	4.82	0.067	0.188	**0.000**
$\bar{\sigma}^2$ lags 2–3	0.649	0.951	0.876	+0.077	+0.050	+0.020	9.85	2.87	4.44	**0.000**	0.080	**0.004**
$\bar{\sigma}^2$ lags 4–5	0.590	0.899	0.948	+0.104	+0.046	+0.021	8.94	3.01	4.38	**0.000**	0.063	**0.040**
Shock lags 0–1	0.681	0.936	0.940	+0.084	+0.016	+0.012	9.67	4.42	4.89	**0.013**	0.061	**0.012**
Shock lags 2–4	0.672	0.908	0.948	+0.098	+0.040	+0.027	8.67	3.21	4.10	**0.002**	**0.010**	**0.016**
Exclude 2020–21	0.642	0.992	0.886	+0.140	+0.043	+0.013	6.40	3.13	4.88	**0.021**	0.052	0.637

Table 9: One-at-a-time variations around the canonical specification. Bold p-values reject cross-group equality at the 5% level.

C.3 Asymmetric extrapolation

Pooling across groups and splitting the shock by sign, both components load positively, with positive news carrying roughly twice the weight of negative news (λ^+ = 0.055 versus λ^- = 0.027), significant across cluster choices (N = 20,363, Hansen J p = 0.11). Upside news is read as informative about future earning power; downside news is more often dismissed as transitory, the opposite of what loss aversion would predict (Kahneman and Tversky, 1979). A second reading is mechanical rather than behavioural: the split is at zero while expected wage growth over the sample was positive, so upside changes were on average the larger surprises relative to what a worker expected, and a rule that responds to the size of a squared surprise rather than to its sign would load more on the upside component for that reason alone. A third reading is that pooling confounds the groups' responses; on any of the three, the asymmetry is suggestive rather than established.

C.4 Numeracy and inflation uncertainty

The SCE's numeracy module classifies 47.0 percent of bottom-group respondents as low-numeracy, against 28.4 percent in the middle and 11.1 at the top, so any mechanical link between numeracy and the width of a reported density is graded the same way as the headline. Conditioning on it compresses the level gradient without removing it: on the full matched panel, a broader sample than the estimation panel, the bottom-to-top difference in perceived variance is 10.6 (1.0), falling to 2.4 (0.8, $p = 0.002$) among high-numeracy respondents and rising to 17.0 (2.1) among low-numeracy ones, with high-numeracy group means of 10.7/8.6/8.7. Re-estimating the belief block (4) within high-numeracy respondents reproduces the middle and the top, with priors of 2.35 (0.85) and 4.93 (1.02), at or below their realised values as in the baseline; the bottom retains only 75 estimation observations under the numeracy-module overlap and its cell is degenerate, so whether the bottom's prior excess survives the conditioning cannot be answered at the estimation level. Two readings of the compression are open: numeracy could widen reported densities mechanically, making the conditioned gap the cleaner estimate, or numeracy is part of the capacity to read fragmented pay, in which case conditioning removes part of the reading-noise channel itself and the conditioned gap is a lower bound, the same over-control argument as the occupation effect of C.1. The gradient survives either reading; its size is bracketed by the two.

Inflation uncertainty is the parallel confound: it is itself steeply graded, 27.9/15.2/8.5 across the groups, and controlling for it absorbs about seventy percent of the bottom-to-top wage-risk gap, leaving a wage-specific gradient of +3.09 (0.60) that remains far from zero. The same two readings apply: general uncertainty could widen the reported wage density mechanically, making the residual the cleaner estimate, or the environment that makes pay hard to read makes prices hard to read too, in which case the control over-partials again. A wage-specific gradient survives either reading.

Three checks on the reporting instrument itself separate mechanics from the channel. Respondents at the bottom use more of the ten histogram bins, not fewer, 5.7 on average against 4.9 and 4.6, and their share of one- or two-bin reports is the lowest of the three groups. Restricting to reports that use three or more bins leaves the gradient intact, 23.0/12.3/12.2. And within every group low numeracy widens the reported inflation density by more than the wage density (log-variance gaps of 0.92 against 1.17 at the bottom, 0.47 against 0.76 in the middle, 0.15 against 0.62 at the top), so the numeracy widening is a generic reporting trait, not a wage-specific one. The level gradient is also not specific to the bin-midpoint variance: under the survey's own density-implied variance, a parametric fit in the tradition of Engelberg et al. (2009), the group means are 28.6/16.4/11.3 against the midpoint measure's 20.8/11.2/11.3 on the same reports, a steeper gradient; the middle-top equality of Section 3 is a property of the midpoint measure, while the ordering and the bottom's excess are not.

C.5 Persistence of the realised volatility process

A rational agent tracking a conditional variance that is itself more persistent at the bottom would show the belief-persistence gradient of Table 1 with no belief friction at all. The realised process does not deliver that: on the raw, unsmoothed cell series underlying the shock regressor, the within-cell monthly AR(1) of the squared change is 0.29 at the bottom against 0.36 at the top (0.44 against 0.49 in the national series), and instrumenting the lag to remove finite-cell sampling noise drives the bottom to 0.06 against the top's 0.26; the bottom's quarterly national series is mildly anti-persistent. The mechanical correlation that adjacent squared changes share through the same person-month is larger where cells are smaller, at the bottom, so the uninstrumented numbers overstate the bottom if anything. The greater persistence of the bottom's beliefs is therefore not inherited from its volatility process.

C.6 The realised process: higher-order moments and the stayer conditioning

Two checks bear on the wage process behind Table 2. First, the permanent-transitory model implies $\mathrm{Cov}(\Delta e_t, \Delta e_{t-k}) = 0$ for all $k \geq 2$. At these sample sizes the higher-order autocovariances are statistically distinguishable from zero but economically second order, between 2 and 23 percent of the lag-1 moment that identifies σ^2_θ, and the largest deviations sit at the quarterly lags for the top (+0.99 and +0.41 at lags three and six against a lag-1 moment of −5.99 in this reconstruction), the signature of quarterly bonus delivery, the delivery gradient of Section 7 showing up in the realised process itself. Second, the stayer conditioning: months immediately preceding an employer change carry elevated volatility at the bottom, about 1.6 times other months, but they are one percent of the sample, so excluding them moves σ^2_θ by less than 0.04 $(\%)^2$ in every group. Selective separation would need to hide a fourfold understatement of the bottom's transitory risk to close its prior gap, against an observed elevation of 1.6 times on one percent of months.

D Identification: the labelled share, the filter, and the scorecard

This appendix gives the filter behind the gain, details the measurement of the labelled share and its robustness, reports the scorecard that sets the measured mechanism against the estimated belief moments, and collects the hours evidence behind the reading-noise gradient.

D.1 The filter

The agent's monthly signal is one squared pay reading, $z_t = (\Delta\tilde{y}_t)^2$ with $\tilde{y}$ the netted reading of (5), with mean $\sigma^2_\psi + V_{x,g}$ and, under Gaussianity, variance $2(\sigma^2_\psi + V_{x,g})^2$, where $V_{x,g} = 2\sigma^2_{\theta,\mathrm{perc},g}$ is twice the perceived transitory prior. Treating the permanent variance as a random walk with drift q observed through the attributed signal $\lambda_g z_t$, the steady-state Kalman gain is

$$\mu_g = \frac{\bar{P}_g}{\bar{P}_g + R_g}, \qquad R_g = \lambda_g^2\, 2(\sigma^2_\psi + V_{x,g})^2, \qquad \bar{P}_g = \tfrac{1}{2}\left(q + \sqrt{q^2 + 4qR_g}\right).$$

The attribution applied to this signal is the full-change attribution (7) rescaled to the signal's own mean, $\lambda_g^{\mathrm{rule}} = \sigma^2_\psi V_u/[\bar{S}_m(\sigma^2_\psi + V_{x,g})]$, so that the filter's resting mean is the permanent belief (8); substituting it collapses the observation noise to $R_g = 2(\sigma^2_\psi)^2(V_u/\bar{S}_m)^2$, so the gain depends on the labelled share and not on reading noise, the perceived prior entering the attribution and the signal variance by offsetting factors and cancelling out of the signal-to-noise ratio. The model carries the gain as the estimated moment; the filter serves as the diagnostic on its ordering reported below.

D.2 Measuring the labelled share

The SIPP records, for each job and month, the dollar amounts of bonuses, commissions, tips, and overtime pay alongside total earnings. On the stayer panel of Section 2, same-employer spells of at least six consecutive months over reference years 2013–2023, 2,250,815 person-months across

103,134 persons and 127,184 spells, I measure the share as follows. For a worker i in job spell s, let y_{it} be her total earnings in month t and ℓ_{it} her labelled pay, bonuses and commissions in the main specification, with spell means $\bar{y}_s$ and $\bar{\ell}_s$. I demean each within the spell and scale by spell-mean earnings,

$$d_{it}^{\text{tot}} = \frac{y_{it} - \bar{y}_s}{\bar{y}_s}, \qquad d_{it}^{\text{lab}} = \frac{\ell_{it} - \bar{\ell}_s}{\bar{y}_s},$$

so the level of pay, the base wage and whatever is fixed within the job, drops out and only the month-to-month movement remains, as a fraction of typical pay and comparable across workers of very different earnings. The labelled share is the person-weighted coefficient from regressing the labelled deviation on the total through the origin,

$$\rho_g = \frac{\sum_{(i,t)\in g} w_{it}\, d_{it}^{\text{lab}}\, d_{it}^{\text{tot}}}{\sum_{(i,t)\in g} w_{it}\, \big(d_{it}^{\text{tot}}\big)^2} = \frac{\text{Cov}_w\big(d^{\text{lab}}, d^{\text{tot}}\big)}{\text{Var}_w\big(d^{\text{tot}}\big)},$$

with w_{it} the SIPP person weights and the sum over stayer person-months in group g. Decomposing the total movement into its labelled and unlabelled parts, $d^{\text{tot}} = d^{\text{lab}} + d^{\text{unlab}}$, gives $\rho_g = [\text{Var}_w(d^{\text{lab}}) + \text{Cov}_w(d^{\text{lab}}, d^{\text{unlab}})]/\text{Var}_w(d^{\text{tot}})$, which reduces to the pure variance share $\text{Var}_w(d^{\text{lab}})/\text{Var}_w(d^{\text{tot}})$ when the two parts are uncorrelated; that variance share is the alternative row of Table 10, close to the regression share throughout. The denominator $\text{Var}_w(d^{\text{tot}})$ is the total within-spell variance of pay, which carries the permanent drift accumulated over the spell and any tenure trend on top of the transitory variation, so ρ_g measures the labelled share of all within-spell movement, a conservative proxy for the labelled share of σ_θ^2 alone. The main specification labels bonuses and commissions, the components that arrive flagged on their own line, and leaves tips and overtime in the unlabelled residual, tips because they arrive as scattered cash, overtime because a variable shift count is exactly the illegible delivery of Section 7. Seam months are dropped as in Section 2. Receipt shares in the same records confirm the direction of the delivery gradient: the share of workers receiving a bonus rises from 2.8 to 4.8 to 9.7 percent across the groups in 2018 (2.8/7.1/12.1 in 2021), receipt of any labelled component from 7.9 to 16.5 percent, while tips, the unlabelled kind, fall from 2.4 to 0.9; receipt speaks to direction, not to the level of ρ, and places the middle below the top on every labelled component. Table 10 collects the specifications.

Specification	Bot	Mid	Top
Main: within-spell demeaned, labelled = bonus + commission	0.121	0.178	0.358
Variance share instead of regression share	0.146	0.203	0.373
Within-spell detrended	0.176	0.256	0.488
Imputed component months dropped	0.042	0.085	0.314
Overtime counted as labelled	0.134	0.197	0.370
Annual spell-year cells (different frequency)	0.050	0.060	0.075

Table 10: The measured labelled share ρ_g across specifications, by income group, SIPP stayer panel 2013–2023, person-weighted. The main row is the specification the model carries. The detrended row removes a within-spell linear trend that demeaning would otherwise leave in the unlabelled residual, and bounds the share from above; the imputation row drops person-months in which any pay component is imputed, and bounds it from below. The annual row aggregates to spell-year cells before computing shares and is reported for completeness, not comparability.

Two caveats attach. First, component amounts in the SIPP are partly imputed, and imputation that spreads labelled dollars across months can manufacture labelled variance; dropping every imputed person-month lowers the shares to 0.042/0.085/0.314 but leaves the gradient intact, so I treat that row as the lower edge of the band rather than the preferred estimate, since it also discards genuine labelled months. Second, the annual specification is a different-frequency object: aggregated to spell-year cells, the month-to-month variation the filter operates on has largely averaged out, and near-annual components such as bonuses contribute little variance within a year, so the annual share measures the labelled fraction of year-to-year risk, not of the monthly signal the belief recursion reads, and it does not enter the band.

A last check concerns the conventions the floor mixes: the share is measured on within-spell variance in levels, while the realised transitory variance of Section 4 is a trimmed difference moment. Measured directly as the lag-1 autocovariance of the *unlabelled* pay change, the floor is insensitive to the trim, moving by less than ten percent in every group where the corresponding total moves by factors of seven to fifty, because the trimmed tail is almost entirely labelled bonus and commission months, the part the mechanism nets out. The labelled share implied by the difference moments, 0.08/0.10/0.48, lies inside the measurement band at the bottom and the middle and at its detrended upper edge at the top, so the band rather than the point is what the top's floor comparison supports, consistent with reading it as directional.

D.3 The scorecard

With ρ_g measured, the estimated belief moments of Section 6 become out-of-sample predictions, and three comparisons score them; a fourth number, the bottom's residual, is what the floors isolate rather than predict. First, the netting floors $(1-\rho_g)\sigma_\theta^2 = 2.21/3.63/3.74$ place the middle and top priors on their floors: the residuals $\sigma^2_{\theta,\text{perc},g} - (1-\rho_g)\sigma_\theta^2 = \hat{\omega}_g^2$ of (6) are -0.68 and $+0.38$ there, with standard errors 0.94 and 0.82, each within one standard error of zero; equivalently, netting predicts shortfalls from realised transitory risk of $\rho_g\sigma_\theta^2 = 0.79$ and 2.09 against estimated 1.46 and 1.71, the same moment stated from the other side. The middle's estimate sits 0.7 standard errors *below* its floor, an insignificant violation of the mechanism's one testable inequality, $\omega^2 \geq 0$, and the model carries the censored value; the violation grows to about 1.2 standard errors at the band's lower edge, so the floor reading is sharpest at the main specification. The bottom's residual, +7.43 (standard error 1.76, $t = 4.2$), is not a prediction the floors could have failed: it is the identification of the reading noise, the excess the floor isolates, concentrated where hours volatility is highest (D.5 below). At the reconciled shares the model carries (Section 9) the floors are 2.22/3.55/3.68 and the residuals +7.43/−0.59/+0.44, the same picture. Classical reporting error in SIPP earnings loads one-for-one into measured σ_θ^2 and hence into the floors, so it can only raise the residuals: at an error share s of measured transitory variance each residual rises by $s\,\sigma_\theta^2$, the middle's stays within one standard error of zero for s up to 0.37 and insignificant up to 0.57, while the top's crosses one standard error at $s = 0.08$ and significance at 0.21; the bottom's excess only grows. Second, the closed form (7) at the measured shares predicts bottom-to-top and middle-to-top attribution ratios of 1.69 and 1.31 (in logs 0.52 and 0.27), against estimated ratios of 3.04 and 1.76 (in logs 1.11 and 0.57, with log standard errors 0.63 and 0.59); the gaps are $z = +0.9$ and +0.5, within one standard error, and the predictions are about half of the estimated point gradient. The moment has little power against the full-information alternative: the FIRE gradient $\lambda^*_{\text{Bot}}/\lambda^*_{\text{Top}} = 1.43$ (log 0.36) sits $z = 1.2$ from the estimate, so the ratios discipline the mechanism's direction and rough size, not its existence against full information. Third, the gain diagnostic below predicts the gains' pinned contrast, the bottom slowest, and underpredicts their spread. The tally is then two moment sets consistent within one standard error, with their power stated, one ordering hit with a spread miss on the moments the cadence comparison of Section 3 marks as the least reliable in the belief block, and one significant residual the mechanism names rather than predicts.

Attribution: ratios, not the level. The cell-mean regressor attenuates the estimated attribution: the cell mean tracks only the common component of a worker's shock, so the slope is scaled down by the share of the squared-shock variance that is common. Treated as common across groups, the factor cancels in the cross-group attribution ratios, which are the moments the scorecard compares; the level is not comparable. The common-factor restriction is an identifying assumption of the merge, not something the data verify. The structural attribution at the measured shares, 0.677/0.524/0.401, stands against the regression level 0.079/0.046/0.026, an implied attenuation of roughly an order of magnitude; that the implied factor is not quite uniform across groups is the same statement as the scorecard's half-explained ratio gradient. The two-percent between-cell share of the squared shock in SIPP makes an attenuation of this order plausible, without pinning it.

D.4 The gain diagnostic

The gain is carried, not derived, but the filter above disciplines its ordering. At the measured shares the observation noise is $R_g = 2(\sigma^2_\psi)^2(V_u/\bar{S}_m)^2 = 262.7/229.2/168.3$, falling with the labelled share, so any common drift q implies a gain rising with income, which is the estimated ordering. At the drift that comes closest to the three estimated gains, $q^* = 646.5$, the predicted gains are 0.763/0.783/0.823 against the estimated 0.629/0.910/0.950: the pinned contrast, the bottom below the middle and the top, is right and the spread underpredicted. The diagnostic carries no weight anywhere in the calibration, since the model reads the estimated gains directly, and the gain levels are in any case the moments Section 3 shows the two survey cadences cannot jointly pin.

The gains also separate delivery from experience-based learning, which grades updating by lived history and hence by age (Malmendier and Nagel, 2011). Within the 40-plus age band the estimated gain rises from 0.57 to 0.84 to 0.98 across the income groups, a bottom-to-top gap of over four standard errors, so the income gradient in the gain is not an age-composition artefact; the under-40 band retains too few bottom-group respondents (117) to estimate.

D.5 Hours by income group

Both forces operate through hours: fragmented, variable hours generate the reading-noise force that lifts the bottom's prior, and transitory shocks delivered through hours are of the unlabelled kind and lower ρ_g. Within-person hours volatility $\mathbb{E}[(\Delta \log h)^2]$ falls monotonically with income (4.10/3.06/2.36 in $(\%)^2$), consistent with the reading-noise force being highest at the bottom (Figure 10). The hours proxy is a one-directional check on it: it supports a reading-noise channel that falls with income, the gradient of the residual above the netting floor (Section 7), while the priors higher up sit on their measured floors (D.3 above). The gradient is directional only: a proportional scaling of the residual in hours volatility, calibrated at the bottom (7.43/4.10), would predict a middle residual near 5.5 against an estimated −0.68 (standard error 0.94), so no intensity mapping from observables to ω^2 is estimable from three group means, two of them zero.

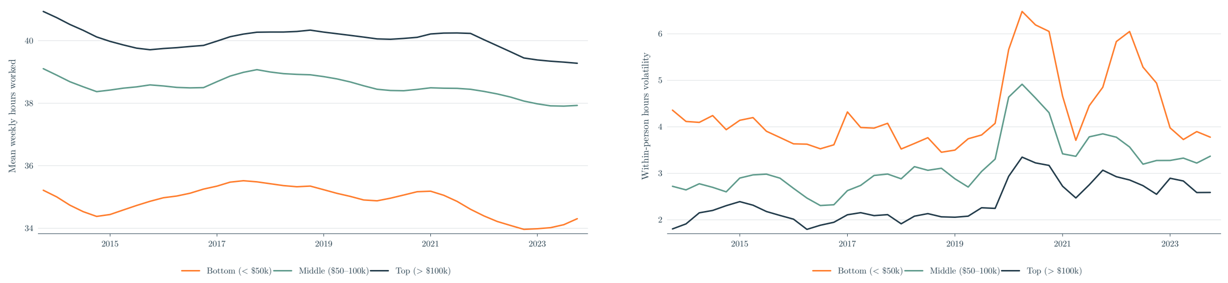

Figure 10: Mean weekly hours (left) and within-person hours volatility $\mathbb{E}[(\Delta \log h)^2 \mid g, t]$ in $(\%)^2$ (right) by income group. SIPP matched-stayer panel, quarterly with 4-quarter trailing MA.

E Calibrating the belief profile

This appendix builds the continuous income-to-belief profile the model consumes from the three estimated groups, and defines the upper-bound belief-anchored variant that Section 10 reports.

E.1 From three groups to a continuum

The continuous profiles of Figure 5 interpolate three income groups; finer bins cannot anchor the belief moments. The SCE elicits income in brackets whose extremes are thin, and splitting the terciles into the five native brackets and re-estimating breaks the belief block exactly where it is split: the lowest bracket (under $20k, 102 respondents) retains no cell that clears the estimation floor, the attribution turns negative in the lower-middle bracket, and the gain exceeds one (an inadmissible negative persistence) in the top two brackets; only the middle bracket, which is the unchanged tercile, survives. System GMM with lagged instruments and two-way clustering needs hundreds of within-person spells per cell, and the bracketed extremes do not supply them. The terciles are the finest split the survey powers, and the continuum is an interpolation through them, not an estimate.

Interpolation. The income profiles in Figure 5 interpolate the three groups with monotone cubic interpolants, the measured share and the carried gain in levels and the carried prior in logs, each held flat outside the anchor incomes so that no belief object is extrapolated beyond its measured range; the tangents at the end anchors are set to zero, so each profile joins its flat extension smoothly rather than at a kink, and the interior tangent carries the Fritsch–Carlson bound that keeps every segment monotone. The prior's interpolant is additionally censored below at the local netting floor $(1 - \rho(y))\, \sigma_\theta^2(y)$, since the mechanism admits no prior beneath it; with the flat tangent at the middle anchor the constraint never binds between anchors, the carried curve touching its floor only at the middle itself. The attribution, the discount, and the permanent belief follow from the closed forms at each income. With three anchors the interpolants set only the shape between groups. Table 11 evaluates the resulting mapping at representative incomes.

Monthly	Measured	Carried		Attribution		Discount	Perceived
income	ρ	μ	$\sigma^2_{\theta,\text{perc}}$	λ	λ^{rule}	$V_u/\bar{S}_m$	perm. var
\$1,500	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$2,415 (Bot)	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$3,500	0.134	0.718	6.72	0.610	0.445	0.952	11.21
\$6,014 (Mid)	0.197	0.910	3.55	0.519	0.568	0.915	10.61
\$9,000	0.297	0.945	3.82	0.453	0.515	0.860	9.81
\$13,604 (Top)	0.369	0.950	4.12	0.398	0.468	0.812	9.11
\$20,000	0.369	0.950	4.12	0.398	0.468	0.812	9.11

Table 11: The income-to-belief mapping of Figure 5 evaluated at representative incomes. The reconciled share ρ and the carried gain μ are interpolated through the three group anchors (the Bot, Mid and Top rows) with monotone cubic interpolants in levels, and the carried prior $\sigma^2_{\theta,\text{perc}}$ in logs, all held flat outside the anchors with smooth joins, and the prior censored below at its netting floor between them (the constraint never binds). The attribution λ is the closed form (7) at each income; λ^{rule} is the recursion's gain, the attribution rescaled to the agent's own perceived signal so that the resting mean equals the perceived permanent variance; the discount $V_u/\bar{S}_m$ and the perceived permanent variance $\sigma^2_\psi(V_u/\bar{S}_m)$ follow from the same identities. Rows away from the three groups are interpolated, not separately measured. In $(\%)^2$ except the dimensionless shares.

E.2 The upper-bound variant

Section 10 reports a variant that takes the estimated attribution responses at face value instead of the measured delivery. It anchors the bottom's attribution at its full-information share $\lambda^* = \sigma^2_\psi/\bar{S}_m = 0.702$ and lets the estimated attribution ratios supply the middle and the top, giving $\lambda = 0.702/0.409/0.231$; the resting permanent belief $\lambda\,\bar{S}_m$ is then 11.89/8.36/5.29, discounts of 1.00/0.72/0.47 on the truth against 0.96/0.91/0.81 under the measured shares, with the prior carried as in the baseline. The variant is an upper bound on the mechanism: it attributes the whole of the estimated attribution gradient to delivery-graded under-reaction, so the gap between its wedge and the measured calibration's measures what rides on netting beyond the measured pay components, or on sampling noise in the belief estimates.

F Steady-state model output

Table 12 gives the perceived variance under the measured beliefs against the full-information benchmark, by income group.

Group	$\hat{\sigma}^2_\psi$	$\sigma^2_{\theta,\text{perc}}$	Perceived $\bar{\sigma}^2_{\text{perc}}$	FIRE	Ratio
Bot	11.3	8.3	153	148	1.03
Mid	10.8	5.1	139	148	0.94
Top	9.7	3.9	124	147	0.84

Table 12: Stationary perceived variance against the FIRE benchmark, in $(\%)^2$ at the 12-month horizon, simulated cross-sectional means by income tercile. $\hat{\sigma}^2_\psi$ is the mean permanent-variance belief, against realised tercile means of 11.8/11.6/11.4, and $\sigma^2_{\theta,\text{perc}}$ the mean transitory prior, against realised 2.9/4.0/5.3, the crossing of Section 7 in the simulated cross-section; $\bar{\sigma}^2_{\text{perc}} = 12\hat{\sigma}^2_\psi + 2\sigma^2_{\theta,\text{perc}}$; FIRE $= 12\sigma^2_\psi + 2\sigma^2_\theta$. Tercile means mix types; the bottom tercile's total sits mildly above FIRE, its prior excess outweighing a small permanent discount.

Table 13 gives the numbers behind Figure 7: mean liquid wealth and the 12-month MPC, belief regime versus FIRE.

	Mean wealth (months)			12-month MPC		
Group	FIRE	Belief	Δ	FIRE	Belief	Δ
Bot	2.28	2.17	−5.1%	0.470	0.477	+0.7pp
Mid	1.71	1.57	−8.3%	0.411	0.423	+1.3pp
Top	1.50	1.29	−13.8%	0.377	0.400	+2.4pp
Workers only, re-terciled among workers						
Bot	2.20	2.07	−5.8%	—	—	—
Mid	1.54	1.39	−9.7%	—	—	—
Top	1.47	1.25	−15.1%	—	—	—

Table 13: Belief regime versus FIRE, by income tercile, under the single calibrated discount factor and mobility. Wealth in months of permanent income; the twelve-month MPC is the constant-monthly-rate annualisation $1-(1-m)^{12}$ of the one-month response m to a small transitory windfall, not a forward-simulated cumulative path, common-random-numbers paired. PE, $r = 2$ percent, $\beta = 0.917$ common (calibrated to mean liquid wealth ≈ 1.8 months), so the wealth and MPC levels are in the data's range and the wedge in Δ is the object of interest. The bottom current-income tercile is 54 percent retirees, the shares falling to 22 and 12 percent above; the workers-only panel restricts to working households, re-terciled among workers, the MPC not recomputed for the split.

Table 14 gives the general-equilibrium counterpart of Section 10: the FIRE and belief economies cleared at a common productivity normaliser pinned from FIRE at a three-percent target rate.

Regime	r (% ann.)	K	Wealth Gini	H2M share
FIRE	3.00	3.65	0.641	0.338
Belief	3.21	3.45	0.645	0.342

Table 14: General-equilibrium FIRE versus belief, common $\beta = 0.917$ (the calibrated value), Z pinned from FIRE at $r^* = 3\%$ and held fixed across regimes. The belief economy under-perceives permanent risk, supplies 5.5% less capital, and clears at a rate 0.21pp higher with a wealth Gini 0.004 higher and a slightly higher hand-to-mouth share. One-asset closure; a two-asset extension is left to future work.